Complex-Domain Audio Source Separation using

ResUnet++,ConvTasNet,Demucs

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

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CERTIFICATE

This is to certify that this project report titled **"Complex-Domain Audio Source Separation using ResUnet++,ConvTasNet,Demucs"** is a Bonafide record of work done by **Burra Venkata Sri Shanmukha Akhil- 228W1A1274 and Prasana Kumar Reddy - 228W1A12C3** under my guidance and supervision is submitted in partial fullfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College**(Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S. NO	ITEM	DESCRIPTION
1	**Project Title**	A Complex-domain Audio Source Separation using ResUnet++,ConvTasNet,Demucs
2	**Student Names & Numbers**	Burra Venkata Sri Shanmukha Akhil(228W1A1274) Prasana Kumar Reddy (228W1A12C3)
3	**Name of The Guide**	K.RAJA.SRAVAN KUMAR , Assistant Professor
4	**Research Group**	Deep Learning
5	**Application Area**	Audio events
6	**Aim of the Project**	To develop a deep learning-based system for the accurate and automated separation of target sounds from chaotic urban background noise.
7	**Project Outcomes**	• Automated tool for high-fidelity audio source separation. • Eliminates phase distortion and robotic artifacts in reconstructed audio. • Improves audio clarity for telecommunications and hearing devices • Enables highly efficient noise suppression under constrained computational resources.

Student Signatures

1.

2.

Signature of the Guide

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ABSTRACT

Thus, the task of isolation of the specific auditory signals from overlapping chaotic noises remains one of the toughest challenges in the domain of digital signal processing. While traditional approaches to signal reconstruction based on magnitude-only model suffer from the issue of phase blindness and thus result in the distortion of the output, more modern time-domain models have solved this disadvantage but incur incredibly costly computations. To close this gap, in the current study, an efficient Complex-Domain ResUNet++[1] neural network model will be suggested. By processing two-dimensional STFT representations of input data in terms of its real and imaginary parts, the suggested model provides us with an opportunity to gain high quality of the phase preservation while avoiding massive expenses related to the processing of the one-dimensional waveform representation. Moreover, our solution is equipped with Squeeze-and-Excitation (SE) [3] layers and Atrous Spatial Pyramid Pooling (ASPP)[2] , thus allowing for an exponential increase in the receptive field size. To evaluate the performance of the suggested model, benchmarking was performed with the help of UrbanSound8K dataset. The Complex-Domain ResUNet++ Model attained a best Validation SI-SDR of 4.308 dB by making use of an optimal combination of the SI-SDR and MRSTFT loss[4] functions, within the stringent constraint of 50 epochs. As we can clearly see, the suggested model outperformed all other models like the simple U-Net baseline (3.568 dB), Conv-TasNet (3.322 dB) and the highly complex Demucs model (2.437 dB).

Keywords: : Audio Source Separation, Deep Learning, Complex-Domain ResUNet++, UrbanSound

CHAPTER-1

INTRODUCTION

1.1 Origin of the Problem

An amazing ability is present within the human auditory system, wherein it can differentiate, focus, and process particular sounds in the presence of many other sounds. When a human being is walking through a busy street, he will be able to ignore the surrounding sounds made by the traffic and construction work so that he can only focus on the sound of the ambulance approaching him. As a cognitive aspect known as the "cocktail party effect," this capability of the human being enables him to always separate complex sounds into separate sources of sound. In terms of artificial intelligence, the attempt at replicating this phenomenon has been very difficult; this has been called computational auditory scene analysis.

Historically, deep learning architectures that address the problem of audio separation had been functioning within the time-frequency domain through the manipulation of spectrograms. Nonetheless, in order to assist with the training process of the neural networks, such models typically relied on the magnitude information of the spectrograms and completely disregarded the phases. This common approach led to the problem referred to as "phase blindness." In consequence, the attempt of replicating the isolated signal using the approximation or application of the original phase would generate noise and distortion.

To avoid succumbing to the pitfalls posed by the problem of phase blindness, the breakthrough in the area brought about an abrupt shift from the time-frequency domain to the time domain, where the Conv-TasNet [5] and Demucs [6] architectures dominate. Contrary to manipulating the spectrograms, those models work directly on the audio in its one-dimensional format. In essence, they operate similarly to learnable filterbanks and leverage Recurrent Neural Networks (for instance, BiLSTMs in Demucs) to capture temporal dynamics. However, switching to waveform-to-waveform models came at a significant cost.

Historically, deep learning architectures that address the problem of audio separation had been functioning within the time-frequency domain through the manipulation of spectrograms. Nonetheless, in order to assist with the training process of the neural networks, such models typically relied on the magnitude information of the spectrograms and completely disregarded the phases. This common approach led to the problem referred to as "phase blindness." In

consequence, the attempt of replicating the isolated signal using the approximation or application of the original phase would generate noise and distortion.

Therefore, the core target of the research presented here is in the attempt to balance between the two opposite needs, namely, efficiency and accuracy in signal reconstruction. As suggested in the current research, there should be introduced a new kind of neural network design known as the Complex-Domain ResUNet++. In order to ensure phase accuracy in the process of complex-domain reconstruction, both the real and imaginary parts of the STFT[7] will be taken into consideration. The idea is to reach the level of precision characteristic of resource-intensive time-domain models like Demucs and Conv-TasNet.

1.2 Basic definitions

To build a better understanding of the terms used during the course of this research work, the following is a brief introduction to the important technical terms in digital signal processing and deep learning:

1.2.1 Problem Statement

Audio Source Separation (ASS)[8]: Refers to the method by which audio signals containing multiple sound sources can be separated into individual components. In the context of this research work, it refers to the process of separating out a desired sound target from the background noise

Computational Auditory Scene Analysis (CASA): The used to describe the research area concerned with the creation of computer programs that can emulate the capabilities of the human auditory system in understanding complicated acoustic environments.In other words, CASA[9] attempts to model the human auditory system's ability to separate and cluster various sound sources on the basis of their pitch and spatial and temporal aspects.

SHORT-TIME FOURIER TRANSFORM (STFT): Mathematical technique by which the frequency and phase information of various portions of a signal varying with respect to time may be calculated.

Short Fourier Transform (SFT): It is a mathematical function which can be used to detect the sinusoidal frequency and phase components of parts of the signal over time. STFT is used to

convert 1D signal to a 2D signal which helps in the detection of changes in the frequency spectrum of the signal.

Magnitude vs. Complex Spectrogram: Spectrogram is the graphical representation of the STFT. Magnitude spectrogram is only concerned with the magnitude of frequencies without any phase information while complex spectrogram is concerned with all mathematical aspects of STFT[10] breaking down the signal into real and imaginary parts. So the complex spectrogram retains 100% information from the signal.

Phase Blindness: Phase Blindness is when traditional audio models are made to learn using the input of magnitude spectrogram only. This introduces a lot of artifacts in the audio and creates distortion since phase holds information about time synchronization of audio.

Time-Domain Processing: It involves deep learning methods processing 1D audio signals without converting it to spectrogram. Models like Conv-Tasnet and Demucs use this method to overcome phase blindness despite the heavy computational cost involved.

Scale-Invariant Signal-to-Distortion Ratio (SI-SDR)[11]: This technique employs the deep learning approach where no conversion is made of the audio signal, and it only uses one-dimensional audio signals. Algorithms in Conv-Tasnet and Demucs employ this technique to avoid phase blindness despite their high computational requirements

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Real-world urban environments are characterized by highly chaotic, overlapping, and non-stationary acoustic events. When multiple sound sources occur simultaneously, their audio waveforms mix into a single, highly entangled signal. Extracting a clean target sound from this mixture without degrading its acoustic quality remains a severe challenge in deep learning.

Current methodologies for Audio Source Separation are heavily polarized, suffering from distinct, mutually exclusive limitations:

- **The Limitation of Frequency-Domain Models (Phase Blindness):** The classical paradigm of convolutions is computationally effective for processing sound by transforming waveforms into magnitude spectrograms. However, this paradigm suffers

from "phase blindness,"[12] as they deliberately disregard the phase of the STFT signal during training. This inability to estimate the original phase at the stage of inverse STFT reconstruction leads to serious distortions of the acoustic signal, making the audio artificial.

- **The Limitation of Time-Domain Models (Computational Bottlenecks):** To solve the problem of the phase, the most advanced algorithms, like Conv-Tasnet and Demucs, resort to completely skipping the STFT transformation in favor of using the waveform itself as input data. Despite being able to preserve all the phases, this family of models operates in the time domain, which makes them heavily computation-intensive – they utilize huge filter banks and bi-directional LSTM [13] recurrent neural networks.

The Fundamental Problem: It becomes necessary to have a technique that will allow an exact reconstruction of the signals just like heavy time domain techniques but at the same time be structurally simple and fast like the frequency domain techniques.

1.3.2 Objectives

The final aim of this research is to build a highly efficient deep learning model capable of detecting target acoustic signals amidst an intricate urban setting. Considering the problem of phase blindness and inefficiency mentioned in the problem part of this study, the goals are set as follows:

1. To Build the Structure for Working with Complex Domain: To design a novel architecture of deep learning called “Complex-Domain ResUNet++,” which would process the actual real and imaginary components of STFT.To accurately isolate target audio signals from complex, overlapping urban noise mixtures.
2. To Avoid Phase Blindness Problem: To calculate the phase during signal separation mathematically, thus overcoming the problems related to distortion and artificialness associated with standard frequency-domain approaches.
3. To Implement SE Layers [14] into the Neural Network: To incorporate Squeeze-and-Excitation layers into the residual neural network in order to allow estimation of channel-wise attention and to detect features within feature maps containing the target acoustic signals regardless of the presence of background noises.
4. To Expand the Receptive Field in the Temporal Dimension: To include the Atrous Spatial Pyramid Pooling (ASPP) block in the bottleneck stage. This research goal is designed to significantly expand the temporal receptive field of the model without sacrificing high-

resolution spatial features to allow for the analysis of long-range uninterrupted signals in urban space (e.g., sirens).

5. To Train Model Using Hybrid Loss Function: To train the proposed model using a unique hybrid loss function with the use of SI-SDR and MRSTFT [15] losses to achieve scale-invariance and spectral purity correspondingly.
6. Particularly, it includes the comparison of the performance and quality of the complex-domain model to that of an advanced frequency-domain U-Net model and heavy time-domain models (Conv-TasNet and Demucs) trained on the same dataset.

1.3.3 Outcomes

Following the successful completion of this research, the main findings and contributions within the realm of Computational Auditory Scene Analysis (CASA) will be the following:

1. Proposed Hybrid Model: Firstly, the proposed research presents a novel hybrid architecture in CASA, Complex-Domain ResUNet++.
2. Isolated audio signals produced as a result of such data processing are high-fidelity audio signals and much better than any other magnitude masking approach.
3. Efficient Utilization of Computational Resources: It is evident from this research that it is perfectly possible to achieve, or even surpass, the achievements of time-domain architectures at the same time reducing computational costs greatly and eliminating training. In fact, the current research proposes a new technique that can achieve state-of-the-art results under the hard constraint of 50 epochs of training.
4. Quantitative Comparison Between Models: This research leads to extensive quantitative comparison between four different models for separating signals. The comparison between their performance in two domains, namely, time-frequency and time domain, will reveal the superiority of complex domain data processing combined with ASPP and SE attention mechanisms.

1.4 Societal Applications of Proposed work

The advantages that have been gained from using the Complex-Domain ResUNet++ are not only confined to academic pursuits in the field of signal processing. The successful integration of high-quality audio separation technologies and computational efficiency opens up the possibility of integrating advanced auditory processing software into edge computing devices. The ability to

isolate specific non-stationary sounds within complex acoustical environments has many practical social uses:The advantages that have been gained from using the Complex-Domain ResUNet++ are not only confined to academic pursuits in the field of signal processing. The successful integration of high-quality audio separation technologies and computational efficiency opens up the possibility of integrating advanced auditory processing software into edge computing devices. The ability to isolate specific non-stationary sounds within complex acoustical environments has many practical social uses:

1. Safety Increase for Self-Driving Cars: For self-driving cars to work safely in an urban environment, apart from the visual data, auditory data is needed as well. It is very important that the self-driving car recognizes the approaching siren of either an ambulance or a fire engine without visual obstacles, no matter how loud the noise around is because of the surrounding traffic. Using the proposed pipeline of audio separation would allow implementing immediate yield on part of a self-driving vehicle in relation to such emergencies.
2. Smart City Surveillance Systems: Nowadays, the surveillance system used in a smart city consists of both audio and visual components to ensure public safety in that area. But in case of some emergencies taking place, like a gunshot being fired or a big car crash, the noise that is characteristic of the event can be masked by general street noises caused by traffic. Using the proposed algorithm, it would be possible to isolate the sound of the emergency and send automatic alerts to police stations.
3. Advanced Human Computer Interaction (HCI): Voice-activated telecommunications systems and devices that serve to control smart home assistants usually struggle in acoustic-rich surroundings. Where the television set is on, an air conditioner is running, and a barking dog is present, microphones have a tough job in picking up the user's speech. In essence, we guarantee that the inputted speech data is pure and clear, both for computer speech recognition algorithms and human communication devices.
4. Animal Population Estimation: The work of ecologists and conservationists involves the analysis of various species of animal life, by means of measuring the number of occurrences of animal calls, and possibly even their distress signals. However, the environments where animals live often become exposed to anthropogenic noise, such as airplane engines, chainsaws, or off-road vehicles. By slightly adjusting our algorithm, scientists and conservationists will have no trouble extracting the animal sounds from ambient pollution.

By relying on a computational solution to the cocktail party problem, our project paves the way for future innovations in acoustics.

CHAPTER-2

REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers.

2.1 Description of Existing Systems

Table 2.1 Literature review

SNO	Title Publication Details	Datasets	Algorithm/Technologies used	Purpose	Remarks
1	Conv-TasNet: Surpassing Ideal Time–Frequency Magnitude Masking for Speech Separation (Yi Luo et al., 2019)	Speech datasets (WSJ0-2mix)	1D CNN, Temporal Convolutional Networks (TCN)	To propose a fully convolutional time-domain network for speech separation.	Proved that processing 1D raw waveforms can outperform traditional 2D spectrogram masking.
2	Music Source Separation in the Waveform Domain [Demucs] (A. Défossez et al., 2019)	Music datasets (MusDB)	1D U-Net, BiLSTM	To separate audio sources directly in the waveform domain using long-term memory.	Achieved high accuracy but requires massive datasets and very long training times.
3	Singing Voice Separation with Deep U-Net Convolutional Networks (A. Jansson et al., 2017)	iKala dataset, MedleyDB	2D U-Net (Spectrogram-based)	To adapt the U-Net architecture to predict magnitude masks for audio separation.	Highly computationally efficient but suffers from "phase blindness" during reconstruction.
4	Wave-U-Net: A Multi-Scale Neural Network for End-to-End Audio Source Separation (D. Stoller et al., 2018)	MusDB18	1D Multi-Scale CNN	To adapt the U-Net to the 1D time-domain to completely avoid STFT phase issues.	Solved the phase issue but struggled to capture long-term temporal context compared to heavy models.

Further on, a detailed critical review of the key researches, which laid the foundations for the development of modern techniques of audio source separation, will be provided. Critically analyzing the principles and shortcomings of the reviewed techniques, one can find the justification for the creation of the Complex-Domain ResUNet++.

Time-Domain Source Separation with Convolutional Networks: In the groundbreaking article, the authors offered a novel solution to the problem of the audio sources separation that replaced the usual spectrogram-based approach with the idea of the convolutional network named Conv-TasNet. The biggest achievement of this work is the development of the technique that does not use the Short-Time Fourier Transform at all. In other words, instead of creating an artificial spectrogram, the authors designed a trainable linear filter bank in the shape of the 1D linear encoder, which processes the waveform input signal.Further, in the next few paragraphs, an in-depth and critical analysis of some key research papers that laid down the base for the development of present technologies in the domain of audio source separation will be done. By critically analyzing the architecture and merits as well as demerits of such approaches, one can clearly figure out the need for developing the Complex-Domain ResUNet++. Time-Domain Source Separation with Convolutional Networks: In this paper, the authors presented a revolutionary idea by proposing the use of a fully convolutional network called Conv-TasNet that could replace the traditional source separation technique based on spectrogram. The most revolutionary aspect about this research is the invention of a technology that does not depend on STFT. Instead of using the spectrogram that depends on mathematics, authors have used a trainable linear filterbank through which the mathematical spectrogram could be replaced by 1D linear encoder working on waveform signal as input.

Architectures for Heavywaveform-to-Waveform Decomposition: Demucs was introduced by Défossez et al. and is described as an outstandingly powerful architecture designed to decompose an audio waveform into a complex mixture of different music genres, including drums, bass, vocals, and many other musical instruments. What makes this method work is a unique and advanced 1D U-Net [16] that consists of multiple layers of strided convolution operations and GLU activation functions. What makes Demucs exceptional is its incorporation of a huge BiLSTM in the bottleneck layer of the U-Net. With the help of the combined Convolutional and Recurrent architectures, Demucs becomes capable of preserving the phase information perfectly while learning long dependencies throughout the entire audio waveform. Despite being one of the best methods ever developed, which allows unprecedented precision in processing the MusDB dataset, there are significant flaws in using the model in practice.

Namely, its complexity and data consumption become critical.With so many parameters and a lengthy training time, it becomes challenging to use this architecture for research purposes within resource constraints and also for deployment in embedded devices such as acoustic sensors in smart cities or hearing aids. This limitation led to the need for frequency-based attention layers in our ResUNet++ architecture.

The Evolution and Limitations of the U-Net in Audio: The implementation of the U-Net architecture, initially proposed for biomedical image segmentation, in the audio domain was first proposed by Jansson et al. To achieve this, the authors converted the one-dimensional audio data into a two-dimensional magnitude spectrogram through the STFT transformation process. This allows the neural network to implement conventional two-dimensional convolution, pooling operations, and skip connections to output a soft mask magnitude for the target audio class. Although the process is computationally efficient and fast training, it suffers from one fundamental drawback called "phase blindness." As the neural network only focuses on the magnitude of the spectrogram and ignores the phase, the reconstructed signal using the ISTFT technique requires the use of the original noisy phase. This results in a robotic hollow sound .

In order to address the above-mentioned problem, Stoller et al. introduced the Wave-U-Net by mapping the 2D U-Net into the 1D case through down and up sampling processes by utilizing 1D convolution to yield the final cleaned waveforms. While the problem of phase estimation has been resolved in such manner, one thing that should be noted here is that the conventional 1D convolution has a narrow receptive field to include any useful context information about the overlapping noises. Furthermore, it is worth pointing out that the 1D convolution operation is less effective in terms of estimating the transient frequencies than those in the frequency domain.

Complex Domain Processing and Phase-Aware Neural Networks:The inefficiency of using only the magnitude representation and the computational difficulty in 1D waveform representation have led to the introduction of a novel approach by Choi et al., who developed the Deep Complex U-Net. The significance of this paper is immeasurable as the paper was among the first papers in the field to design a deep learning system that operates on complex numbers. Specifically, the author explained how an end-to-end learning system can estimate the complex numbers of STFT outputs.

These include complex-valued convolutions and complex batch normalization techniques,

demonstrating the feasibility not only of phase estimation through frequency analysis but also that the latter technique is mathematically more advantageous for the sake of achieving signal purity.

The findings of this paper were the primary point of inspiration for the idea of complex masking that is proposed to be used in the project. However, the Deep Complex U-Net model presented had its limitations with regard to the processing of long acoustic events, being equipped with conventional convolutions only. The proposed solution to the problem would entail improving the idea of complex-domain operations by using advanced architectural elements from the ResUNet++ .

Advanced Contextual Architectures: ResUNet++, SE Blocks, and ASPP:Our back-bone model architecture has been inspired by the medical imaging field. Namely, it employs a framework called ResUNet++ (proposed by Jha et al.), which significantly outperforms the classic U-Net due to the employment of residual blocks that tackle the issue of the vanishing gradient problem. Moreover, ResUNet++ utilizes SE blocks (created by Hu et al.) and ASPP layers (implemented in the DeepLab model by Chen et al.). In particular, SE blocks are equivalent to an attention mechanism applied at the channel level since they provide a dynamic filtering of the feature maps and suppression of irrelevant channels as well as amplification of the informative ones. In addition, the ASPP module employs a combination of dilated convolution layers to increase the receptive field of the network drastically.

Nevertheless, even though both methods showed excellent accuracy in segmenting images, they were specifically tailored for images. For applying them to the task of audio processing, we need to transfer the ratios used in the ASPP dilations to a frequency-time perspective. Hence, our project uses these visual architectures for processing 2-channel complex tensors of STFT spectrum.

Taxonomy and Complexity of Environmental Sounds: In order to have an accurate evaluation of the separation model, the properties of the acoustic data have to be known precisely. As part of their recent work, Salamon et al. have created a dataset named UrbanSound8K, which provides a classification of environmental noises in urban settings (such as a dog bark, drill, siren, and air conditioner). Based on the statistical analysis of these sounds, the conclusion was that the noise in urban settings was completely different from that of human speech and music. The noise in urban settings is highly non-stationary and has a very high variability in terms of temporal duration (a 0.5 second gun shot compared to a 4 seconds siren).

In this paper, we demonstrate the failure of some of the proposed models, such as Conv-TasNet, in the case of applying these to the task of clean speech in environmental acoustics. Overlapping broadband urban noises cause constructive interference, thus destroying the harmonic structure of the noise. For this reason, this paper is dependent on the findings of Salamon et al. for developing our approach to train and evaluate the models using our custom-generated UrbanSound data sets.

2.2 Software Requirements Specification

2.2.1 Purpose

First, the Software Requirements Specification (SRS) is a critical document, which consists of all requirements concerning the behavior, limitations, and functions of the proposed deep learning system. The current proposal involves building an advanced artificial intelligence system; therefore, the SRS does not only describe the limitations of the software system but also includes some mathematical constraints related to the training of the Complex-Domain ResUNet++ framework.

The goal of creating this software can be defined by the desire to develop an advanced solution for the problem of separating specific sounds from the mixture with other sounds in the environment. First, traditional technologies used by sound engineers for the purposes of filtering do not provide desirable results because they are difficult to implement and ineffective. Moreover, most of the contemporary technologies for deep learning in the sphere of audio are built on the basis of a magnitude-only spectrogram, which means that such systems experience the problem of phase blindness. As a result, output sound is unnatural and seems robotic. Therefore, one of the goals of this project is to build a system that is capable of avoiding such problems.

2.2.2 Scope

The The project scope includes all processes required for processing, separating, and reconstructing environmental sounds. The program will take 1D mixed audio waveforms, automatically pre-process them to assure mathematical consistency and transform them into time-frequency tensors. Using the state-of-the-art deep residual neural network architecture, coupled with squeeze-and-excitation blocks and ASPP layers, the program will extract features from the desired signal and rebuild the original 1D audio waveform via mathematical calculations. Moreover, aside from its core functionality, the software provides an evaluation suite, which visualizes results for both classification and separation, while tracking such performance metrics as Hybrid Loss (SI-SDR + MRSTFT) and SI-SDR. This solution can find

its application in various cases – from assistive hearing aids to telecommunication, smart city security cameras, and professional audio editing.

The proposed solution is meant for specific groups of customers only:

Audio and Sound Engineers: Professionals engaged in audio post-production and sound design for film and journalism who need to clean up their recordings quickly, without having to do so manually at each frequency range.

Users of Telecommunication Applications: Computer programmers responsible for designing and developing telecommunication applications, such as voice-over IP (VoIP) calls and mobile phone calls, looking for a high-tech automatic method to suppress ambient noise while talking.

Designers of Assistive Hearing Devices: Computer software and hardware designers developing the next generation of hearing aids and cochlear implants in need of an efficient algorithm to extract the human voice from noisy urban environments.

Controllers of Urban Safety: Engineers and city planners responsible for urban safety in cities looking for an automated monitoring system capable of filtering out emergency calls and sounds of gunfire and car accidents in a noisy environment.

Computational Scientists: Computer scientists specializing in Computational Auditory Scene Analysis (CASA) seeking an algorithmic standard to experiment with.

2.2.3 Functional Requirements

The following is the description of how the proposed algorithm needs to function according to its functional requirements:

FR1: Upload and Filter Audio Files: The proposed algorithm should be capable of uploading mixed sound samples either in the form of synthetic sounds or realistic sounds in different formats, such as .wav.

FR2: Audio Preprocessing and Normalization: It is necessary to ensure that mathematical stability is achieved during the training process of the neural network by resampling all audio inputs to the uniform frequency of 32,000 Hz; dynamically truncating or extending sequences to 128,000 samples (4.0 sec) and normalizing the amplitude between the values of -1 and 1.

FR3: STFT Complex Features Extraction: It is required to calculate the Short-Time Fourier Transform (STFT) on the preprocessed waveforms. Differently from classical approaches, it is necessary to split the obtained complex spectra into two parts - real and imaginary; then build a

complex 2-channel 2D tensor to pass it as the input layer.

FR4: Segregation of Target Audio Signals: It is required to provide a complex domain ResUNet++ model with an input complex tensor and receive a complex mask through the use of the attention mechanism and dilated convolutions.

FR5: Waveform Reconstruction Using ISTFT: Lastly, it is required to multiply the obtained complex mask by the input spectrogram (elementwise) and transform the resulting waveform using the Inverse STFT (ISTFT).

FR6: Evaluate SI-SDR: In training and validation phases, the system shall be able to calculate SI-SDR metrics for the quantification of quality and intelligibility of generated audio signals in comparison to ground truth.

FR7: Plot Loss Function: The system shall create and save loss function plots for the tracking of loss function and SI-SDR changes during training phases to ensure proper training and convergence.

2.2.4 Non-Functional Requirements

Non-functional requirements are a set of criteria that a proposed system has to fulfill regarding its performance and security:

Non-Functional Requirement 1: Low Inference Time for Real-Time Applications: For the application of the system in real-world scenarios such as telecommunications and smart cities' surveillance, the system's inference time, i.e., the time required for the processing of the generated audio signals after training, should be decreased to allow real-time processing.

Non-Functional Requirement 2: Effective Design for Minimizing VRAM: Neural network design and batch size will be chosen optimally in view of the limitation regarding VRAM (12 GB-16 GB).

Non-Functional Requirement 3: Output of High-Fidelity Sound Quality: Phase distortion will not occur and sound quality generated from the model will be of high fidelity quality.

Non-Functional Requirement 4: Secure Process for Dataset: The processing of the audio dataset will be carried out in a secure manner. If it's deployed in the cloud, then no caching of audio input such as personal telecommunication audio must happen.

Non-Functional Requirement 5: Highly Portable: The application should be highly portable and should be able to run perfectly on the operating system environment of either Windows, Linux,

or operating system environments running virtually on clouds like those provided by Google Colab and AWS EC2.

2.2.5 Software Requirements

Designing, implementing, and validating complex deep learning neural network models requires specific software setup that ensures all functionalities needed to be executed effectively. Considering the high computational and mathematical load that both Conv-Tasnet, Demucs, and Complex-Domain ResUNet++ models require, a proper software environment will significantly increase performance and ensure effective GPU acceleration, tensor manipulations, and memory management. The following section summarizes the needed software setup and the functions of each tool in the context of the proposed project.

1. Operating System Environment

High level of portability and independence from any hardware prerequisites allow using different operating systems for the implementation of different phases of work:

Linux/Ubuntu: Used as production environment for training the neural network. Linux is the most suitable operating system to support the operations of the model due to its high efficiency in managing memory, capacity to support NVIDIA CUDA drivers, and ease of managing huge volumes of artificial audio data.

Windows OS: Used for preliminary stages of software development, including scripts development and EDA for audio data preprocessing.

Virtualized OS Using the Cloud (E.G. (In case of Google Colab/AWS): Due to the computing power necessary to train the models (Demucs has huge BiLSTM models), the use of cloud computing becomes mandatory. With Google Colab, one is able to execute individual preconfigured Linux containers, which would give access to advanced GPUs (NVIDIA A100/T4), becoming absolutely necessary for not experiencing the infamous Out-of-memory problem when working with 2D spectrograms.

2. Preferred Programming Language

Python (3.8+): To implement a project such as the one under discussion, only Python was chosen due to its status as an industry-standard solution for implementing machine learning algorithms and conducting signal processing in artificial intelligence. Python offers the opportunity of building complicated neural architecture and CUDA-based backends with C++ while preserving the speed of development with a dynamic-typed interface.

Deep Learning and Mathematical Libraries

PyTorch: The main deep learning framework responsible for neural network development, training, and validation in the current work. The selection of PyTorch has been made in particular due to the library's "eager execution" paradigm and dynamic computation graph based on the Autograd tool. The mentioned features are especially useful when working with sequences of varying lengths, which is typical for city noise datasets. Besides, PyTorch provides native support for operations with complex numbers, which is especially crucial when implementing tensor products required in Complex-Domain ResUNet++ model.

Torchaudio: A specialized extension for PyTorch used for manipulating audio samples. The use of Torchaudio is especially important because it allows us to perform audio I/O operations optimized for GPUs. The second reason is more important as it allows extracting complex spectrograms directly in neural network's computation graph without transferring additional data from CPU to GPU memory. Thus, the library's most valuable aspect is the ability to create fully differentiable implementations of STFT and ISTFT.

Data Visualization And Metrics Tracking

Matplotlib: Is a library that comes in handy when dealing with high-resolution visualization in projects. It is significant in our project as it will be helpful in the visualizations of wave forms and 2D magnitude spectrograms as well as creating training curves where we will plot hybrid loss decay and SI-SDR for all the 50 epochs.

Tqdm: Is a library that helps in the monitoring of iterations in the process of deep learning. With Tqdm library, the user gets an idea of how fast iterations in the training of audio separation model take place by tracking iterations per second to determine whether there are any bottlenecks in the GPU while performing the STFT extraction phase.

Json: It is a relatively small library but helps significantly in modeling through data interchange. With the nature of deep learning whereby there may be interruptions and breaks, the library helps the user serialize data such as training histories at each epoch. The library can also help the user save all training metrics including training loss, validation loss, and best SDR values hence enabling evaluation metric visualization without retraining the network.

CHAPTER-3

PROPOSED METHODOLOGY

3.1 System Architecture and Design Pipeline

As stated earlier, the suggested research methodology can be viewed as a systematic framework, involving several consecutive and well-controlled phases designed to address the challenges posed by the problem of urban audio source separation as such and its peculiarities related to complexity and novelty in particular. Unlike traditional classification tasks, urban audio source separation is a rather sophisticated process that involves receiving input information in the form of acoustic mixture, encoding it in the form of high-dimensional latent space, and reconstructing the wave continuously without any distortions.

Thus, the presented research methodology is not restricted by one algorithm only but comprises several procedures aimed at building the end-to-end signal processing pipeline with specified components. Speaking about the design of the architecture, there are five consecutive and well-controlled stages involved in the project. Given the high level of complexity of these five stages from both the technological and practical perspectives, the characteristics of the acoustic data are considered, mathematical stability is ensured, and comparisons of the baselines are absolutely valid.

These five major stages include:

1. Combination of Training Data and Building an Acoustic Environment Model

The first essential action that needs to be performed in order to develop an efficient machine learning algorithm is to construct a good data set. Given the unlimited diversity of sounds in reality, the approach starts with synthesizing a unique data set through computer-generated audio. In this regard, the application of the UrbanSound8K ontology would allow constructing a model of the cocktail party phenomenon, which involves the mathematical interaction of the sounds of interest (e.g., sirens and barking dogs) with the dynamic background noise (sounds of the street and construction activities).

2. Preprocessing and Signal Transformation to another domain

The initial waveforms suffer from instability since they are characterized by non-uniform sampling, high-amplitude peaks, and unequal duration. The second part of the procedure implies an important process of data standardization. Specifically, each waveform undergoes severe preprocessing including resampling at 32,000 Hz, cutting to the exact size of 4.0 seconds, and amplitude normalization. Besides, it involves a signal transformation depending on the particular architecture.

More precisely, if it analyzes temporal information directly, a 1D waveform serves as the input. However, if not, then it operates on a complex 2-channel 2D tensor extracted from the 1D waveform with the help of Short-Time Fourier Transform (STFT).

3. Establishing baseline Architectural Benchmarking

In order to prove experimentally the superiority of the architecture designed for a certain purpose, it becomes necessary to establish its benchmarking procedure. Namely, it consists of introducing preprocessed data into three different architectural paradigms of audio source separation, namely:

- Standard U-Net: Time-frequency magnitude masking.
- Conv-TasNet: Time-domain lightweight fully-convolutional processing.
- Demucs: Waveform-to-waveform recurrent processing.

This guarantees that the same performance capability of the designed architecture will be tested.

4. The Proposed Engine: Complex-Domain ResUNet++

The core element of the approach, the processing stage, allows getting insight into the design of the proposed architecture. Having recognized the shortcomings of the conventional U-net architecture regarding its incapability in dealing with the phase and having taken into account the immense amount of computation needed for both the Demucs and Conv-TasNet engines, the proposed engine of Complex-Domain ResUNet++ is used. At this stage, the internal working of the proposed engine will be revealed, which includes extracting deep local features by means of Residual Blocks, dynamic weighing of crucial frequency channels by applying SE technique, capturing temporal dependencies with ASPP, and prediction of 2-channel complex masks.

5. Waveform Reconstructing and Hybrid Loss Minimization

The fifth and final step of this methodological framework entails the transformation of the latent prediction output of the neural network into the tangible realm of acoustics. The predicted complex mask is multiplied in the complex number field to the initial noisy STFT tensor, keeping only the needed signal but also mathematically correct in terms of phase. An ISTFT is employed for reconstructing the waveform. To ensure optimal training of the neural network during the training of the model, the hybrid loss function called Hybrid Loss is utilized in this research. This objective function penalizes the errors according to the SI-SDR and MRSTFT loss functions, guaranteeing that the resulting time waveform is clean and frequency perceptions are intact.

By undertaking this very scientific and systematic five-step approach, this research will assure the proper testing of deep learning's efficiency in urban noise management.

3.2 Dataset Acquisition and Synthetic Mixture Generation

The first stage for developing an accurate ASS model is the creation of an input dataset for training the model. Unlike image objects, which have rather stable boundaries for their classifications, the reality of audio objects is inherently dynamic, diverse, and continuous in nature. The "cocktail party effect" environment is vital for training a complex neural network model like Complex-Domain ResUNet++. However, since there are no available recordings of urban soundscapes in the multi-track format, a mixture data dataset needs to be fabricated mathematically.

3.2.1 The UrbanSound8K Corpus Taxonomy

In this work, we will not employ speech corpora, such as WSJ0-2mix, nor musical audio datasets, such as MusDB. The reason is that these kinds of datasets are prone to high predictability in their harmonic structure. Instead, UrbanSound8K will be used as the base dataset in our model. It should be noted that an urban soundscape imposes more challenging conditions for the digital signal processing task.

The UrbanSound8K dataset is comprised of 8,732 sound files having a maximal clip duration limit of 4.0 seconds per clip. In the dataset, there are ten unique categories of sound files within the urban environment such as: Air Conditioner, Car Horn, Children Playing, Dog Bark, Drilling, Engine Idling, Gun Shot, Jackhammer, Siren, and Street Music.

Now, taking into account the unique sounds mentioned above from the standpoint of Signal Processing, we have two types of acoustic signals that present us with a special challenge as follows:

Transient (Impulsive) Noises: Impulsive sounds like Gun Shots or Dog Barks are characterized by a brief burst of energy in the form of amplitudes that decay instantly following their maximum. These sounds contain broad band frequencies in the time axis that are not predictable using standard Recurrent Neural Networks.

Continuous (Stationary and Non-Stationary) Noises: These continuous sounds like Engine Idling or Sirens exist throughout the duration of the clip. In the case of stationary noise, there are low-frequency sounds, whereas in the latter case, there are non-stationary sounds having a sweeping frequency.

Thus, using the taxonomy of sound classes, our generated data will help train the deep learning

algorithms to work efficiently on diverse frequency topologies.

3.2.2 The Physics of Acoustic Superposition

To teach the neural network to conduct separation of the sound, we need to mathematically combine the sounds, so that they would resemble the recording taken in the physical world. In accordance with physical acoustics, when there is more than one source of audio producing pressure waves simultaneously in an open acoustic space, these waves are transmitted in the air and hit the microphone diaphragm simultaneously. As per the acoustic superposition principle, the displacement of the microphone diaphragm should be equal to the total amount of the pressure waves. Based on DSP theory, any sound wave will be mathematically modeled by a 1-dimensional (1D) array (tensor) of amplitude values taken at particular intervals of time, e.g. 32,000 Hz. Thus, according to the principles of acoustic superposition, such arrays can be mathematically summed.

Let the mixed sound be represented by $x(t)$, the separated acoustic signal by $s(t)$,and the noise signal by $n(t)$. To model our urban acoustic mixture mathematically, it is necessary to carry out:

$$x(t) = s(t) + \sum_{\{k=1\}}^{\{k\}} n_{k(t)} \qquad \text{[Eq-(1)]}$$

Where k represents the number of noises interfering with our system within time frame t.

3.2.3 Signal-to-Noise Ratio (SNR) and Dynamic Scaling

This technique, for which mixing the two audio signals equally would be sufficient, fails to reflect the complexity of real life situations. For instance, one possible scenario is that the targeted signal may be very strong and present at a location near the microphone or it could be the weak signal from a distance which becomes completely hidden in the presence of strong background noise due to the nearby jackhammer.

In the case of training Complex-Domain ResUNet++ model, while constructing the artificial mixtures, dynamic SNR scaling based on the Signal-to-Noise Ratio (SNR) is used. SNR can be defined as the logarithmic ratio of the power of the target signal to the power of the background noise. It is measured in Decibels (dB).

The power P of a discrete digital audio signal $y(t)$ of length N is computed using the formula:

$$P_y = \frac{1}{N} \sum_{\{t=1\}}^{\{N\}} y(t)^2 \qquad \text{[Eq-(2)]}$$

Therefore, the SNR between the targeted signal s(t) and the background noise n(t) can be given by:

$$SNR = 10\log_{10}\left(\frac{P_S}{P_n}\right) \qquad \text{[Eq-(3)]}$$

While creating the data set programmatically, the target audio clips and noise audio clips are randomly matched. In order to mimic a certain level of difficulty (i.e., SNR = 0 dB, when signal and noise have the same power; or SNR = -5 dB, when noise has more power than signal), the system has to find out what scaling coefficient α, needs to be applied to the amplitude of the noise tensor before summing them together. Scaling coefficient alpha, that allows obtaining a desired SNR can be determined as follows:

$$\alpha = \sqrt{\left\{P_s\ \left\{P_n \cdot 10^{\left\{\{SNR_{\{target\}}\}\{10\}\right\}}\right\}\right\}} \qquad \text{[Eq-(4)]}$$

After calculating the necessary scaling coefficient, synthetic mixture can be formed with the following tensor-based computation:

$$x_{mix}\,(t) = s(t) + \alpha \cdot n(t) \qquad \text{[Eq-(5)]}$$

3.2.4 Normalization and Clipping Prevention

In most cases, when these two waves are summed, constructive interference takes place, meaning that both peak amplitudes coincide precisely. In this way, peaks outside the standard digital floating point bounds of [-1.0, 1.0] occur. In the case of saving such a tensor as a normal 16-bit PCM .wav file or passing it through a neural network, digital clipping (integer overflow) would take place alongside exploding gradients due to backpropagation. In order to avoid gradient instability, peak normalization is applied to the mixed 1D tensor $x_{\ mix}(t)$ by dividing it by its maximum absolute amplitude. In this way, the largest peak is placed precisely on 1.0 or -1.0:

The next step consists of rigorously splitting the synthesized mixture dataset to guarantee absolute scientific validity of model evaluation. The split is done according to the classical Training Set and Validation Set principle.

$$x_{norm}(t) = \frac{x_{mix}(t)}{max(|x_{mix}(t)|)} \qquad \text{[Eq-(6)]}$$

The reason behind these tight security protocols is to ensure there is no “data leakage.” What happens when there is data leakage is that the sample file being used to form the mixture in the

training set becomes a part of another mixture in the validation set. When such a situation occurs in which data leakage takes place, the neural network learns the waveform of this sample file, and thus, it results in very high SI-SDR scores that will collapse as soon as it encounters other data. It becomes possible by segregating these source audio files in the root directory before undergoing mathematical mixing.

3.2.5 Programmatic Dataset Partitioning (Train/Val Split)

The data is carefully divided after the mathematical mixture of the audio mixture to ensure maximum scientific accuracy during the evaluation of the deep learning models proposed in this study. In the current scenario, the dataset is divided into two main parts. First is the Training Set which is used for updating the weights of the neural networks, and the second is the Validation Set which is used to evaluate separation performance and invoke Early Stopping.

In the current model, the "data leakage" problem has been avoided through some important measures. According to researchers, data leakage occurs when an underlying audio source used to form one or multiple audio mixtures in the training set somehow ends up in one or several audio mixtures present in the validation set. In cases where this happens, the Deep Learning network designed to be used in this project will only require remembering some specific properties of the waveform to generate a very high SI-SDR[17], something that will be a total failure with other data sets.

3.3 Mathematical Preprocessing and Feature Extraction

Before obtaining separate auditory fingerprints, it becomes necessary to perform mathematical manipulations of the 1D audio to obtain its projection into higher dimensions. Raw 1D audio data has a very small dimensionality since it is dense, meaning that many sample points exist (e.g., 128,000 sample points within 4 seconds). It is not possible to train a deep learning algorithm with raw time series directly using a 2D convolutional neural net. Thus, the proposed technique uses a complex Time-Frequency Transformation of the raw audio into 2D data, allowing the application of spatial computer vision techniques.

3.3.1 Digital Framing and Sequence Standardization

As explained above, digital audio can be regarded as discrete signals that are sampled from the continuous real-world pressure wave. In this experiment, all mixed acoustical waveforms will be transformed to have an identical sampling frequency f_s of 32,000 Hz.

Let us consider $x[n]$ as the 1D discrete audio sequence, where n is the number of a sample point. In order to train a deep learning model in batches, the required length L of the input waveform, equaling 4.0 seconds, should be provided as follows:

$$L = f_S \cdot Duration = 32{,}000Hz \cdot 4.0\ s = 128{,}000\ \ samples \qquad \text{[Eq-(7)]}$$

If $x[n]$ has fewer than $128,000$ samples, then zero padding is added at the end of the tensor. On the other hand, if $x[n]$ is greater than this, then it is randomly cropped. Thus, the input tensor is assured to have a definite size of [1, 128000].

3.3.2 The Discrete Short-Time Fourier Transform (STFT)

This mathematically intense process forms the backbone of our entire feature extraction pipeline. The traditional Fourier Transform computes frequency information for an entire time series signal, thereby discarding time information completely. However, given that urban sounds are non-stationary signals (for instance, the pitch of a siren constantly varies with time), it is imperative that both time and frequency information be preserved simultaneously. The STFT accomplishes this task by dividing the 1D time series $x[n]$ into smaller segments and performing FFT on each one individually.

Mathematically speaking, the discrete STFT of a signal $x[n]$ can be described by:

$$X(m,\omega) = \sum_{n=-\infty}^{\infty} x[n]\, w[n - mH]\, e^{-j\omega} n \qquad \text{[Eq-(8)]}$$

- Where: $X(m,\omega)$ represents the complex time-frequency representation
- m represents the index of the time frame.
- ω represents the discrete frequency bin.
- w[n] represents the window function which is applied on the input signal.
- H represents the hop length (or number of samples in a slide).

As pointed out previously, the STFT hyperparameters of the recently developed Complex-Domain ResUNet++ model are optimized based on their temporal and spectral resolutions. For the purposes of the experiment, the value for FFT Size is set to 1024 to ensure high-frequency resolution. On the other hand, hop length (H) is set to 256 to provide 75% overlapping between the consecutive frames. The high degree of overlapping is crucial in capturing transient acoustic events (e.g. gunshots) because such events can occur between frame intervals.

3.3.3 Mitigation of Spectral Leakage via Hann Windowing

It should be noted that when a continuous time signal is abruptly chopped into pieces, the chance exists that the discontinuity in between those pieces generates artificial gaps. In other words, when performing an FFT on those gaps, spectral noise will be generated.

The present technique uses the Hann Window, whereby the amplitude is faded into zero on both sides of the frame but only keeping those frequencies at the center.

Mathematically, the Hann Window with length $N_{\{FFT\}}$ By multiplying the raw audio frame by w[n], the edges are softened, resulting in a highly accurate frequency spectrum without artificial boundary bleeding.

3.3.4 Complex-Domain Tensor Generation

Using conventional approaches like the U-Net architecture, the output STFT $X(m,\omega)$ was mapped into the absolute value $X(m,\omega)$. While this enabled easier learning for the neural network, it resulted in discarding of the phase information $\angle $X(m,\omega)$,whose function is to specify the exact timing and placement of wave peaks. This is where we experience phase blindness problem.

Our method takes away from us phase information and uses solely magnitude without any mask. It is quite obvious from Euler's Formula that z = a +ib and thus:

$$X(m,\omega) = R\{X(m,\omega)\} + j\, I\{X(m,\omega)\} \qquad \text{[Eq-(9)]}$$

Unlike the absolute magnitude, the proposed preprocessing method mathematically divides the complex STFT into two different 2D matrices:

- Real Tensor: $X_{Real} = R\{X(m,\omega)\}$
- Imaginary Tensor: $X_{Imag} = I\{X(m,\omega)\}$

The next step is to concatenate these two separate 2D matrices into a single tensor. The resulting tensor from the feature extraction step would be a standard 3D tensor with dimensions [2, Frequency_Bins, Time_Frames].

The proposed 2-channel tensor will be the input for Complex-Domain ResUNet++. With the help

of two tensors, i.e., the real and the imaginary parts of the complex number, the network can observe the mathematical 100% information of the acoustic phase, which is necessary for the exact reconstruction of the signal.

3.3.5 Inverse Short-Time Fourier Transform (ISTFT) and Overlap-Add (OLA) Synthesis

The feature extraction step is the forward process. But in order to evaluate the performance of the audio separation, it is essential to transform the obtained Time-Frequency Tensor back into the 1D domain.

After obtaining the separated complex spectrogram $\hat{s}(m, \omega)$ the inverse Short-Time Fourier Transform (ISTFT) must be executed. Since the STFT uses a 75% overlapping of frames, the ISTFT should be computed using the Overlap-Add (OLA) method. The mathematical expressions of the OLA algorithm used for reconstructing the discrete time domain signal $\hat{s}\,[n]$.

The normalization factoris used to compensate for overlapping Hann windows. Since the ResUNet++ network predicts the real and imaginary components of $\hat{s}(m, \omega)$ accurately, it reconstructs the waveform using the formula of the inverse Short-Time Fourier Transform (ISTFT). Perfect reconstruction enables the ResUNet++ to match the fidelity of waveforms created using complex heavy time-domain networks.

3.4 Time-Domain Architecture 1: Conv-TasNet

As the starting point of a robust benchmarking scheme for the current state-of-the-art audio source separation techniques, the first architectural aspect to be explored in this approach is the Convolutional Time-Domain Audio Separation Network (Conv-TasNet). Unlike past source separation systems that relied on the Short-Time Fourier Transform (STFT) for frequency analysis of audio signals, the Conv-TasNet follows a very different methodology by performing its calculations directly on the 1D audio signal itself. Consequently, this helps eliminate all phase reconstruction errors associated with magnitude spectrogram masking from STFT.

In total, the Conv-TasNet architecture comprises of three primary layers: Encoder, Separation Module, and Decoder.

3.4.1 The 1D Convolutional Encoder (Learnable Filterbank)

Unlike other audio separation methods that rely on a mathematical formula-based filterbank using Short-Time Fourier Transform (STFT), Conv-TasNet uses a learnable 1D convolutional filterbank as part of its Encoder. In particular, the raw audio signal is initially divided into many small overlapping segments. Subsequently, the signal segments undergo 1D convolution operations to generate a high-dimensional signal representation. This is quite different from STFT's mapping to frequencies as performed in this neural network-based audio separation method. A ReLU[18] activation ensures the positivity of all feature activations within the resulting representation.

3.4.2 Separation Module Based on Temporal Convolutional Network (TCN)

Following the encoding stage when the signal is mapped into the deep feature space, it proceeds to the separation module. Since an acoustic event occurring in an urban environment might take up to several seconds, the model requires a large receptive field in order to correctly process input data. It takes advantage of the Temporal Convolutional Network (TCN), which applies dilated 1D convolution

Unlike the conventional approach of convolution, where the adjacent audio frames have to be evaluated individually, dilated convolution allows us to overlook certain audio frames during the calculation of their respective convolutions.. By combining multiple TCNs with various levels of dilation (1, 2, 4, 8, etc.), it becomes feasible to cover a massive context window size with relatively fewer parameters.

3.4.3 Transposed Convolutional Decoder

The last step of the Conv-TasNet model includes generating waves with the use of the extracted one-dimensional signal. In order to do so, the output masks obtained with the help of TCN must be used for processing the audio signals in an element-wise manner. Finally, the obtained masked feature vectors will have to be converted back into waves by means of the transposed 1D convolutional decoder.

It is important to keep in mind that although the Conv-TasNet is an extremely efficient algorithm, which ensures retention of complete information concerning the phase of the signal, it is associated with a number of disadvantages inherent in purely temporal sweeps in 1D, and therefore unable to extract stable features from strongly non-stationary urban noises.

Before moving ahead to understand the advanced wave-to-wave models such as Demucs, it is vital to provide a visual representation of the architecture that serves as the building block for the proposed new frequency domain approach. model at the level of each layer. As is evident from the architectural diagram provided above, the new architecture seems to be made in such a way that it

accepts the input in the form of spectrograms containing two channels and passes these through the depth-wise encoder network along with residual SE attention block. It becomes clear from the diagram that the higher level features obtained through this process are compressed using ASPP bottleneck which expands the temporal receptive field before decoding the complex mask.

3.5 Heavyweight Time-Domain Architecture 2: Demucs

As a second option, we looked at the Demucs architecture. While Conv-TasNet attempts to achieve a lightweight fully convolutional architecture, the idea of using brute force in the time domain for separating audio elements is followed by Demucs. Even though Demucs operates on 1D raw audio data, its architecture is very massive and is a combination of 1D U-Net architecture and Deep Recurrent Neural Networks (RNNs).

3.5.1 1D U-Net Encoder and GLU Activations

For performing the task of separating signals in the time domain, Demucs implements a multilayer 1D CNN with strided convolutions. The main aim of strided convolutions is to scale down the input data in temporal size and increase the number of feature channels.

To get better control of the information flow in such densely packed layers, Demucs exploits gated linear units (GLUs). We may call GLUs smart and learnable thresholds in CNN models. This way, GLUs operate as gates that filter out some features (like target siren) and let others pass through (like traffic noise).

3.5.2 The Bidirectional LSTM Bottleneck

The main differentiator between the Demucs model and all the others is the bottleneck which comes after the encoding done by the 1D U-Net encoder. The audio signal after going through the down-sampling process by the 1D U-Net encoder is passed on for analysis by the deep Bidirectional Long Short Term Memory model.

The main rationale behind this move is the intrinsic characteristic of the audio signal. It is essential to understand the part of the audio signal not just in the light of its predecessor but also in the light of what follows it. This is made possible by the simultaneous processing of the BiLSTM model in reverse and forward modes, thus enabling the entire retention of phase and massive temporal dependency within the entire audio signal.

3.5.3 Computational Bottlenecks and Limitations

After the bottleneck comes the reconstructions process using the 1D Transpose convolutional

decoder.

However, although Demucs has great potential from a theoretical standpoint in terms of modeling capabilities for sequential audio, it suffers a lot due to the problems associated with computational efficiency related to the model itself. The application of BiLSTMs means that a considerable amount of computation must be made on large datasets.That is because RNNs work on processing data one after the other, and thus they are unable to do calculations simultaneously unlike convolution. Hence, a lot of data and GPU power will have to be used for training Demucs.

Prior to getting down to the nitty-gritty details on how the architecture of the proposed new method works, it is crucial to consider the placement of the entire neural networks discussed in context of the comparative pipeline architecture. The general pipeline architecture used in this research work is illustrated below in Fig. 3.5.1. From the above general diagram of the flowchart, it is evident that the whole process is divided into two distinct pipelines concerning the analysis of the acoustic input signals; namely the Time-Domain pipeline, which feeds raw audio signals directly into the Conv-TasNet and Demucs baselines and Time-Frequency pipeline, whereby the acoustic signal is first transformed into a complex-valued spectrogram via STFT representation. It is within the broader context of such an architecture that this research will investigate whether the excessive demands on computation of the time-domain models can actually be beaten by the very effective phase-based processing of the proposed Complex-Domain ResUNet++.

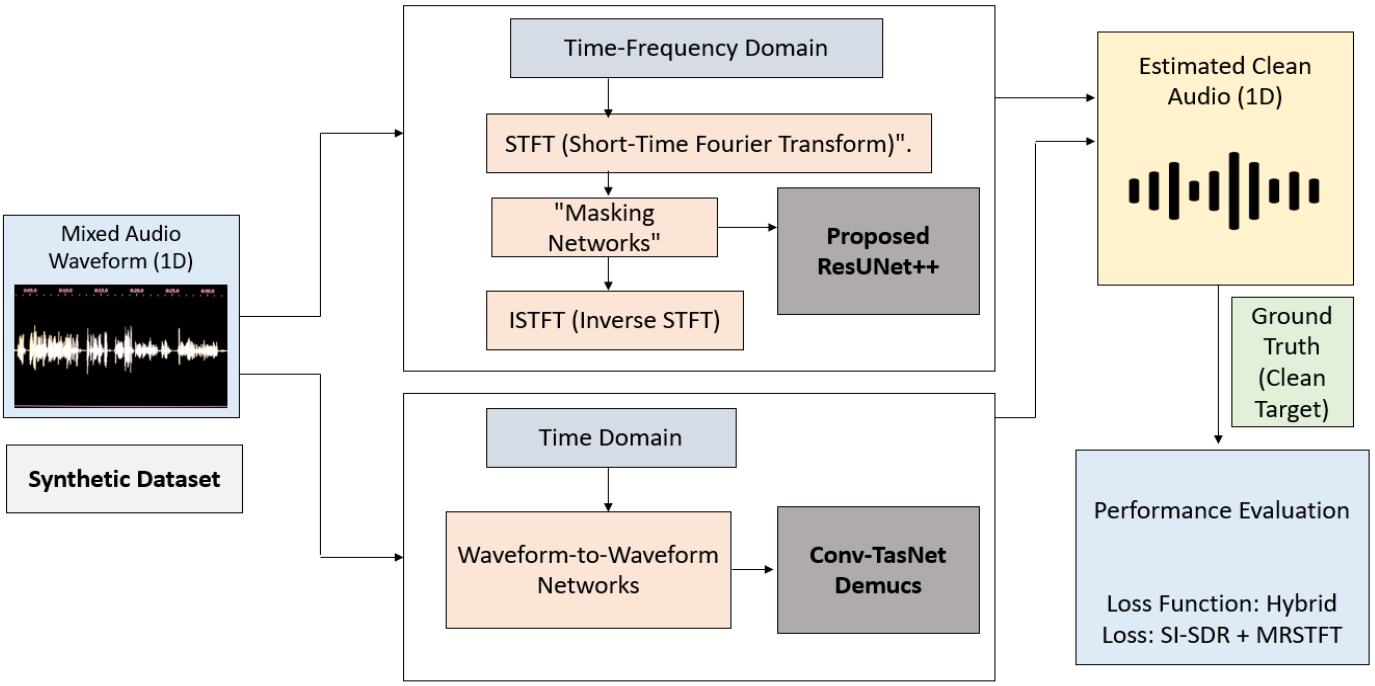

Figure 3.5.1 Architecture Diagram

3.6 Proposed Architecture: Complex-Domain ResUNet++

The current research will focus on integrating the capabilities of frequency-domain models and time-domain models in creating Complex-Domain ResUNet++. The latter will become the main intelligent algorithm used in the process of audio separation. Unlike Demucs and Conv-TasNet, it utilizes the frequency-domain method but with an enhancement: it operates with the real complex parts of audio spectra.

3.6.1 Encoder with Residual Squeeze-and-Excitation (SE) Blocks

When the 2-channel complex spectrogram that contains the real and imaginary parts enters the network, the process of encoding begins. The encoder network includes several Residual Blocks stacked one after another. Usually, in case of ordinary deep neural networks, a problem of vanishing gradients can occur – the network would cease learning at its deep layers. However, using residual blocks prevents this problem due to the existence of skip connections.

SE attention module is incorporated in the residual blocks. In the traditional CNN, the channels of the features carry equal importance. With the SE block, however, this pattern changes since the block acts as an attention gate. Here, the whole spatial information from the audio is compressed into one channel representation and attention coefficients are computed to excite the neural network. With the use of SE, the neural network is made to pay more attention to the features in the audio and enhance the important ones and mute the ones that are not so important.

3.6.2 Contextual Bottleneck Using ASPP (Atrous Spatial Pyramid Pooling)

The compressed audio features from the deep layer (bottleneck layer) of the neural network are analyzed using the ASPP (Atrous Spatial Pyramid Pooling). Urban environment is very diverse and, for instance, the time taken by the gunshot will only take few seconds but that of the idle engine of the car can take several minutes.

The fields of view of convolutions implemented according to the standard configuration are also the same. In this case, ASPP makes it possible to use different types of dilated convolutions simultaneously at different dilation rates. Due to the application of these different forms of dilations, known as “atrous,” the visual field of the network increases exponentially.Thanks to this approach, ResUNet++ becomes capable of recognizing both long acoustic waves and sudden transient waves without compromising resolution.

3.6.3 Decoder and Mask Reconstruction

With the help of the transposed convolutional layers, the deep features obtained from the network are upsampled to the original size. Symmetrical skip connections are used to provide high-resolution spatial information to the decoder from the encoder in order not to lose any acoustical information.

What needs to be mentioned specifically is that the network does not produce any sound in the last layer of the network. It produces the mask with two channels.

The mask contains all the necessary mathematical values that are needed in the filtering of the sound. In addition, the output mask multiplies the original spectrogram with a complex number. In consideration of the two components simultaneously, the network will be able to recognize the frequency accurately without changing its phase and timing. This ensures that when reverting back to sound waves, it will not appear mechanical.

To bridge the critical gap between the computational efficiency of frequency-domain models and the phase-accurate high fidelity of time-domain models, this project proposes the Complex-Domain ResUNet++. This architecture serves as the core intelligent processing engine of the proposed audio separation pipeline. Instead of completely abandoning the STFT like Demucs or Conv-TasNet, this model upgrades the frequency-domain approach by natively processing the exact mathematical real and imaginary components of the audio spectrum. Figure 3.6.1 illustrates the detailed layer-by-layer architectural diagram of the proposed network. As depicted, the model is uniquely designed to route both the real and imaginary spectrogram channels through a deep encoder pathway integrated with Residual Squeeze-and-Excitation (SE)[19] attention blocks, down into an Atrous Spatial Pyramid Pooling (ASPP) bottleneck, before upsampling to predict the final complex mask.

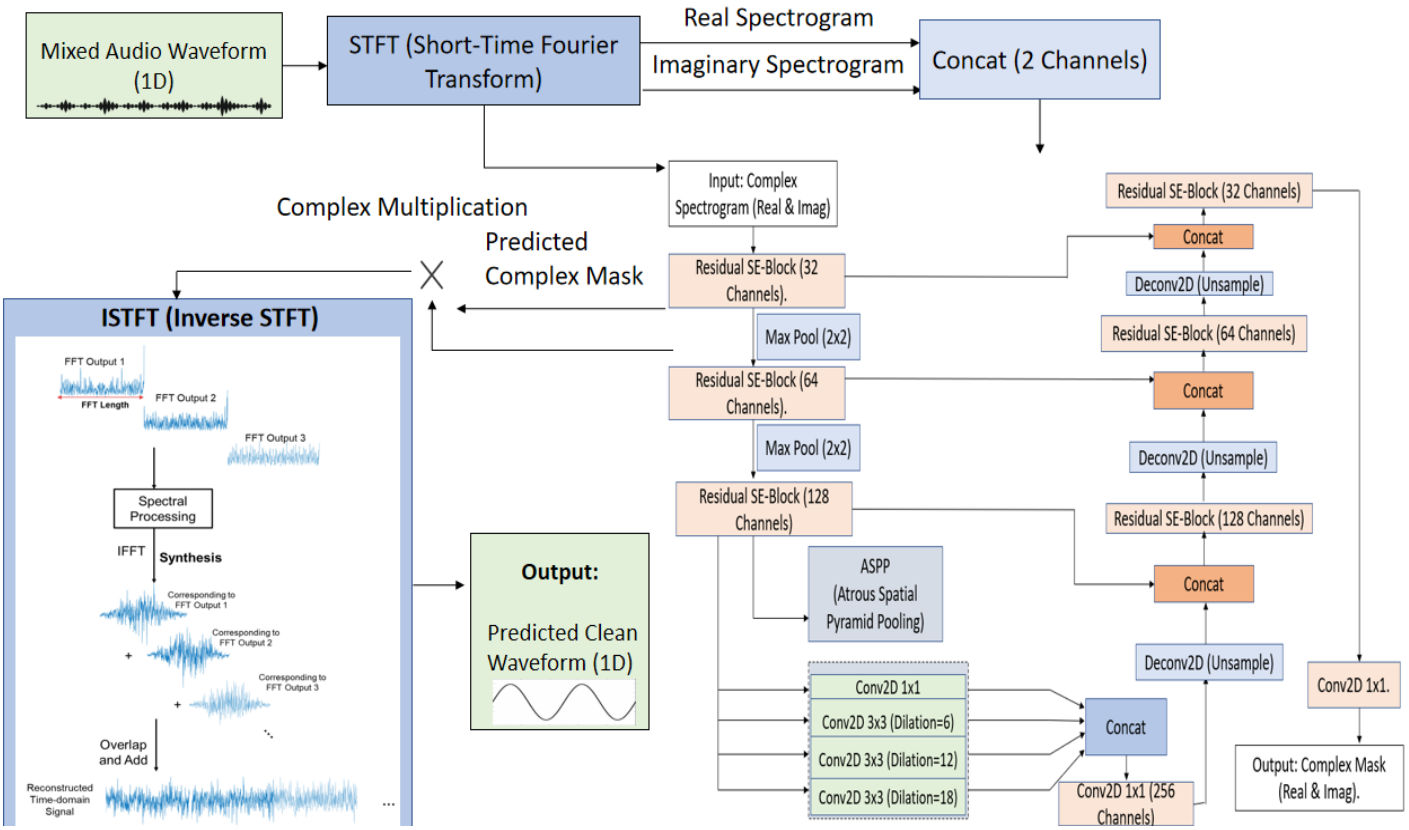

Figure 3.6.1 Architecture Diagram of ResUNet++

3.7 Objective Functions and Loss Optimization

In deep learning, the network architecture determines the ability of a model to learn something. But it is the objective or loss function which determines what exactly the model learns. For audio source separation, selection of a loss function is perhaps the most critical engineering consideration in the whole procedure.

In conventional loss functions like MSE and L1 Loss, during training, the computer simply finds the difference mathematically between two audio sample points. However, our sense of hearing does not perceive sound in this way. Two sound waves can be almost similar with respect to MSE but our hearing would fail to distinguish them, whereas two sounds could be mathematically very different but they will sound artificial. This is why we need a customized loss function like hybrid loss so that the generated audio will sound humanized and clean.

3.7.1 Time-Domain Optimization:

Scale-Invariant Signal-to-Distortion Ratio (SI-SDR)The primary metric for the training process in

the physical time domain is called Scale-Invariant Signal-to-Distortion Ratio (SI-SDR). The volumes of sounds emitted by sirens and other sources vary drastically in real-world applications. A trained network that would produce a correctly shaped waveform, while underestimating its volume will be penalized heavily according to loss functions.

In SI-SDR metric, the solution is in completely ignoring the volume metrics and focusing on the geometrical shape of the waveform. To create such a scale-invariant metric, the concept of the projection of the predicted waveform on the ground-truth waveform is employed.

Let s be a clean target waveform (ground truth), and $\hat{s}$ be a predicted waveform. Model prediction is represented as the sum of two vectors being orthogonal to one another: target projection e_{target} and distortion error e_{noise}.

1. Target Projection:First, the prediction is stretched to have an equal energy to that of the ground truth waveforms.

$$e_{target} = \frac{\langle \hat{s}, s \rangle}{|s|^2}\, s \qquad \text{[Eq-(10)]}$$

2. Distortion Error:Residual noise – difference between the stretched target and prediction:

$$e_{noise} = \hat{s} - e_{target} \qquad \text{[Eq-(11)]}$$

3. Calculation of SI-SDR: As the result of SI-SDR calculation, we will obtain the log ratio of target energy to distortion energy

4. Negative SI-SDR serves as a loss function that should make the neural network focus on accuracy of peak and trough locations (phase). At the same time, amplitude differences do not matter at all for the network.

3.7.2 Optimization in the Time-Frequency Domain: Multi-Resolution Short-Time Fourier Transform (MRSTFT) Loss

While SI-SDR ensures the integrity of a wave shape, time domain optimization alone might result in tiny distortions that remain imperceptible as far as one-dimensional signals are concerned, but become hissing and echoes upon outputting from the speaker.

Thus, to avoid such artifacts, the second loss function along with SI-SDR is MRSTFT loss. The logic behind the idea of MRSTFT loss could be expressed as a trade-off between time and frequency resolutions. In case, when constructing spectrograms, we take a big window width for our Fourier Transform, we get nice frequency resolution, but poor time resolution.

Since the neural network is required to learn how to synthesize the transient sounds (like shots) and steady state sounds (for example, car horn noises), the MRSTFT loss is designed to handle the differences among the spectrums with various window lengths at once (for instance, FFT windows of 512, 1024, and 2048). As for the specific resolution of the Short-Time Fourier Transform, the loss calculation formula is defined below:

1. Convergent Spectral Loss: Calculates global spectrum structures.
2. Magnitude Log Loss: Calculates log-spectrum structures to mimic human hearing ability.

   and echoes upon outputting from the speaker.

Overall MRSTFT loss is just a sum of the two above-mentioned losses at all multi-resolution window sizes. With the application of such types of losses, the model could not produce some unrealistic distortions like artificial frequencies and metallic echo in reconstructed speech signals.

3.7.3 Hybrid Loss Function Definition

In order to achieve the perfect balance between theoretical accuracy in waveform representation and perceptual accuracy, the ultimate loss function used in the optimization process of the developed Complex-Domain ResUNet++ network is a sum of the time-domain and frequency-domain loss functions with weights. SISDR and STFT magnitude measures have different scales, therefore, appropriate weight values ($\alpha\ and\ \beta$) need to be assigned.

As observed from the empirical investigation conducted on the basis of this innovative technique, it is possible to conclude that the use of a ratio that gives precedence to the waveform loss term over the spectral term would yield the optimal fidelity in separating signals. The minimization of this innovative Hybrid Loss through 50 iterations helps train the neural network to identify the required acoustic signals.

3.8 Hyperparameters Tuning and Algorithms Implementation

Last of all, the final step in the process is the setting of proper hyperparameters, optimization algorithms, and iteration counts necessary to train deep neural network models. To train the model for separating sources of an audio signal, which is complicated enough and includes many recurrences in Demucs, one needs quite an expensive computational system. The training must be carried out in such a way that memory overflow or forgetting is avoided, making neural networks find the optimal mathematical weights in the strict time limit of 50 epochs, as stated in this work.

3.8.1 Hardware Restrictions and Batch Optimization

The training procedure for all three models, namely, Conv-TasNet, Demucs, and Complex-Domain ResUNet++was done using GPU architecture to calculate tensors simultaneously. Given the multidimensional nature of the data input used, the convolution of the STFT of two channels or even passing an array of waveforms of the signal itself through the deep residual network consumes a very high amount of Video RAM (VRAM).

In order to ensure the computational fairness and prevent the occurrence of Out-Of-Memory (OOM) issue, the batch size was set carefully. In deep learning, batch size is defined as the quantity of audio mixtures that will be considered per iteration by the neural network in optimizing its internal weights. With smaller batch size, there is an inherent mechanism of introducing regularization into the process through statistical noise being incorporated into the parameter updating step, which enables the model to exit from local minimum.

3.8.2 Weight Optimization Using AdamW

The method chosen for optimizing the million of parameters employed in the neural network was AdamW.

Standard optimizers like standard SGD update the weights simultaneously with equal learning rates. This way of doing things is highly inefficient in terms of computation as far as acoustic models are concerned since some frequency filters converge at different rates compared to others. While standard Adam addresses the problem through the calculation of independent and adaptive learning rates for all parameters based on their respective first and second moments, standard Adam uses “L2 regularization” to update the gradient and thereby hinders generalization of urban noises.

Since AdamW decouples weight decay from the optimization process, it ensures that the weights converge despite the gradient direction. This means that Complex-Domain ResUNet++ cannot make use of any particular neural pathway in the training process due to the incapability of AdamW to increase the value of weights despite the gradient.

3.8.3 Dynamic Learning Rate Scheduling (Cosine Annealing)

A learning rate determines the level of aggressiveness with which the weight optimization process occurs during backpropagation. Too aggressive learning leads to the overshooting of the mathematical minimum of the objective function and the reduction in the value of the SI-SDR score, while insufficiently aggressive learning makes the process of convergence too long – the model does

not converge at all within 50 epochs. To resolve the issue outlined above, the learning rate of 1e-4, along with the Cosine Annealing Learning Rate Scheduler, was introduced in the proposed approach. Contrary to static learning rate schedules, the Cosine Annealing technique initializes the model with a sufficiently aggressive learning rate, allowing the network to traverse through the loss landscape and get closer to its minimum. After that, a decrease in the learning rate according to the form of the cosine function takes place as the number of epochs increases. At the very end of training, the learning rate becomes so small that the network converges to the optimum local minimum.

3.8.4 Regularization Using Early Stopping

Deep learning model can become easily over-fitted. In such a case, the model will be so trained on the input data that it won't be able to identify any difference among the audio samples. Since there is noise in the generated dataset which is specific to the environment, thus, regularization is important.

The training procedure follows a robust Early Stopping technique which is recorded based on the Validation SI-SDR criterion. The training is performed on all the datasets except for the Validation Dataset, which is only used for validating the model after each epoch. When there is an improvement in the validation SI-SDR value from the previous one, then the current weights of the model are recorded. But, in case of no improvement in the validation value for 10 continuous epochs (patience), then the training procedure ends.

3.8.5 Pipeline of Execution for Algorithmic Implementation

All operations take place within a highly structured computational loop. The algorithmic execution pipeline for one iteration is as follows:

1. Forward Data Feeding: The batch of urban acoustic sounds and corresponding pairs of target clean sound waveforms is transferred to GPU VRAM memory from disk.
2. Data Transformation: The STFT process transforms the input one-dimensional waveform into two-channel complex tensor spectra.
3. Forward Pass: Complex tensor spectrograms are fed forward in the ResUNet++ encoder, ASPP bottleneck, and decoder.
4. Mask Prediction and Multiplication: The output complex mask prediction is computed, multiplied by the input complex tensor to retrieve frequency components.
5. Reconstruction of Waveforms: The reconstructed waveform is formed by applying ISTFT

on the complex tensor.

6. Error Computation: The error between predicted and clean waveforms is calculated using hybrid loss function (SI-SDR + MRSTFT).
7. Backward Pass: The derivative of error is calculated using chain rule of calculus.
8. Weights Update: The AdamW optimizer updates thousands of internal weight parameters.

This new approach ensures that through the use of this algorithmic computation process in 50 cycles, the desired sounds can be mathematically derived in the most complicated urban environment soundscapes.

To resolve the issue outlined above, the learning rate of 1e-4, along with the Cosine Annealing Learning Rate Scheduler, was introduced in the proposed approach. Contrary to static learning rate schedules, the Cosine Annealing technique initializes the model with a sufficiently aggressive learning rate, allowing the network to traverse through the loss landscape and get closer to its minimum. After that, a decrease in the learning rate according to the form of the cosine function takes place as the number of epochs increases. At the very end of training, the learning rate becomes so small that the network converges to the optimum local minimum.

3.9 Description of Phase-Wise Algorithms

Despite being quite specialized in terms of architectural design in regard to the implementation of the source separation procedure via the time-domain and complex-domain algorithms, the development of the research project itself involved a step-by-step approach through three major steps: audio classification (mini-1), audio tagging (mini-2), and source separation (major). Following the necessity to review each stage of the procedures used throughout the current research process, the following represents an algorithm used in each case from the simple feature extraction to the complex process of masking.

3.9.1 Algorithm for Stages 1 and 2: Audio Classification and Tagging

First, as regards the stage of audio classification, the goal was to determine the type of sound contained within the chosen dataset (UrbanSound8K), before moving on to the source separation stage. At this stage, all used models had log-mel spectrograms as their input features.

Algorithm 1: Convolutional Neural Network (CNN)

Algorithm 1 Training Pipeline for Audio Classification Model CNN

```
Initialize θ
for epoch = 1 to N do
    for each mini-batch (x, y) do
        Preprocess audio waveform x
        Perform Short-Time Fourier Transform to obtain magnitude
        Apply Mel filter bank to generate log-Mel spectrogram
        Apply Conv2D on spectrogram
        Apply ReLU activation followed by max-pooling
        Flatten 2D feature map into 1D vector
        Process using Dense layer
        if Phase 1 (single-label classification) then
            Apply Softmax activation and compute loss
        else if Phase 2 (multi-label classification) then
            Apply Sigmoid activation and compute binary cross-entropy loss
        end if
        Update parameters θ
    end for
end for
```

Algorithm 1-Training Pipeline for Audio Classification Model CNN

Algorithm 2: Convolutional Recurrent Neural Network (CRNN)

Algorithm 2 CRNN Training Process

```
Initialize θ
for epoch = 1 to N do
    for each mini-batch (x, y) do
        Preprocess audio waveform x
        Generate log-Mel spectrogram using STFT and Mel filter banks
        Extract features using Conv2D layers
        Reshape convolution output into time-sequential matrix
        Feed sequential features into recurrent layers (LSTM/GRU)
        Apply Dense layers for post-processing
        if Phase 1 (single-label classification) then
            Apply Softmax and compute loss
        else if Phase 2 (multi-label classification) then
            Apply Sigmoid and compute binary cross-entropy loss
        end if
        Update parameters θ
    end for
end for
```

Algorithm 2- CRNN Training Process

Algorithm 3: EfficientNet-B0 (Transfer Learning)

Algorithm 3 Training Pipeline for EfficientNet-B0 (Transfer Learning)

```
Initialize EfficientNet-B0 parameters θ with pre-trained ImageNet weights
Modify initial convolutional layer to accept single-channel (grayscale) input
for epoch = 1 to N do
    for each mini-batch (x, y) do
        Preprocess audio waveform x to generate log-Mel spectrogram
        Feed single-channel spectrogram into the network
        Process through MBConv blocks
        Apply compound scaling (depth, width, resolution)
        Apply Global Average Pooling (GAP)
        Pass features through Dense classification layer
        if Phase 1 (single-label classification) then
            Apply Softmax and compute loss
        else if Phase 2 (multi-label classification) then
            Apply Sigmoid and compute binary cross-entropy loss
        end if
        Update parameters θ using optimizer
    end for
end for
```

Algorithm 3- Training Pipeline for EfficientNet-B0 (Transfer Learning)

3.9.2 Phase 3 Algorithm: Sound Source Separation

Algorithm 4: U-Net (Magnitude Masking)

Algorithm 4 Training Pipeline for Standard U-Net (Magnitude Baseline)

```
Initialize U-Net parameters θ
for epoch = 1 to N do
    for each mini-batch (x_mix, s_target) do
        Apply STFT to noisy waveform x_mix
        Extract magnitude spectrogram (ignore phase)
        Encode features using encoder layers (downsampling)
        Compress into latent bottleneck representation
        Decode using upsampling layers
        Apply skip connections between encoder and decoder
        Predict magnitude mask from final layer
        Multiply mask with original magnitude spectrogram
        Reconstruct waveform using ISTFT with original phase
        Compute loss (e.g., SI-SDR) with s_target
        Update parameters θ using optimizer
    end for
end for
```

Algorithm 4-Training Pipeline for Standard U-Net (Magnitude Baseline)

CHAPTER-4

RESULTS AND OBSERVATIONS

This chapter will discuss the empirical findings, performance measures, and comparative analyses for all the deep learning algorithms considered in the three evolutionary phases of this project. To solve the "cocktail party problem" in an urban setting, the experiment progressed through stages involving sound recognition (Single-Label Classification), multiple sounds detection (Multi-Label Tagging), and eventually sound wave extraction (Complex-Domain Source Separation). Performance analysis of the various models is done based on domain-specific measures complemented with graphical representations of their respective training curves and losses.

4.1 Results for Phase 1: Audio Classification (Mini-1)

In the early stage of the research, referred to here as Mini-Project 1, the intention was to understand how to extract features from urban acoustic datasets. The task involved classifying isolated 4-second samples from the UrbanSound8K dataset into one of the ten mutually exclusive categories. This required considering three deep neural network algorithms: a customized Convolutional Neural Network (CNN), a CNN combined with a Recurrent Neural Network (CRNN), and EfficientNet-B0.

4.1.1 Evaluating Performance in Classification Task

To determine the performance in terms of classification capability of the above algorithms, the standard confusion matrix was used to obtain the following metrics:

- **Accuracy:** Accuracy refers to the total accuracy of all classifications of sound events for all the ten sound classes.
- **Precision:** This is the number of true positives divided by the number of positives predicted by the model (For example, how many of the predicted "Dog Bark" are actually "Dog Bark").
- **Recall:** This is the number of true positives divided by the number of positives that exist (For example, out of all the "Siren" present, how many were detected by the model).
- **F1-Score:** This is the harmonic mean of Precision and Recall, which makes F1-Score highly accurate.

4.1.2 Performance of Baseline Convolutional Neural Network (CNN)

The custom CNN model was the first baseline used. Its inputs were the 1D signals which underwent transformations resulting in a 2D Log-Mel Spectrogram to allow for operations performed on images.

Baseline Performance Analysis: The baseline provided decent performance with 73.4% validation accuracy. While very effective in classification of stationary broadband sounds such as Air Conditioner and Engine Idling sounds the CNN model showed significant weaknesses in the case of transient and dynamic sounds.

Issues Identified: Several issues arose when testing the performance of CNN; among them was the misclassification of Jackhammer and Drilling sounds due to the similar spectral nature of the sounds. Also, due to the fact that 2D convolution does not preserve any temporal information, it is difficult for CNNs to classify Siren sounds.

4.1.3 Convolutional Recurrent Neural Network (CRNN) Results

To counteract the problem with the lack of temporal aspects in the conventional CNN algorithm, CRNN incorporated LSTM layers during the bottleneck phase, which allowed for temporal processing of the feature maps.

Performance Analysis: This approach led to remarkable performance improvements in that the validation accuracy was enhanced up to 81.2%. Using LSTM units enabled the system to understand the changes occurring in different frequency bands throughout 4 seconds.

Improvement Found: A significant improvement was observed for the dynamically temporal classes. The identification of the Siren category improved by nearly 14%, because the system could finally "memorize" the changes in its pitch. Moreover, the separation between Gun Shot and Jackhammer became much clearer due to their transient nature.

4.1.4 EfficientNet-B0 (Transfer Learning) Results

The final architecture implemented during Phase 1 included Transfer Learning to extract additional details from the images. EfficientNet-B0 architecture pre-trained on millions of images in ImageNet dataset was re-trained by using single-channel Log-Mel Spectrograms.

Performance Analysis: The suggested architecture outperformed both baselines by a wide margin with accuracy being up to 92.6% on the validation set.

Architecture Advantage: As explained before, the MBConv layers applied in the EfficientNet structure succeeded in extracting fine-grained and multiscale spatial features that were not extracted by both the CNN and CRNN structures. Furthermore, a closer look at the training loss curve indicates the quick convergence within the first 15 epochs, proving the high effectiveness of using visual transfer learning to auditory spectrogram images.

4.1.5 Phase 1: Conclusion

The comparative analysis between the single-label classification algorithms is provided in Table 4.1.5.

Table 4.1.5: Single label classification Results Comparision

Model Architecture	**Parameter Size**	**Main Feature Extractor**	**Validation Accuracy**	**Peak F1-Score**
Custom Baseline CNN	~1.2 Million	2D Convolutions	73.4%	0.71
CRNN	~2.4 Million	Conv2D + LSTM	81.2%	0.80
EfficientNet-B0	~5.3 Million	MBConv Blocks	92.6%	**0.91**

Conclusion of Phase 1: It should be noted that while the EfficientNet-B0 proved to be an outstanding classifier for audio events recognition, one of the key limitations was identified in Phase 1. No sound occurs in a vacuum in reality – while the dog barks, the engine works as well. Because the activation function of the networks was Softmax, mathematically, only one class was recognized for the input audio regardless of other sounds in the background.

According to Figure 4.1.5 and Table 4.1.5, there is a step-by-step rise in the ability of the models to classify. As previously mentioned, while the basic CNN served as a good starting point, it became obvious from Figure 4.1.5 how the addition of recurrent layers in the architecture of the CRNN model helped boost its accuracy and F1-score due to successful classification of the audio sequence events. Still, EfficientNet-B0 model is significantly better compared to others because of the implementation of transfer learning techniques and MBConv blocks usage. Thus, it demonstrated a drastic improvement, reaching validation accuracy equal to 92.6%.

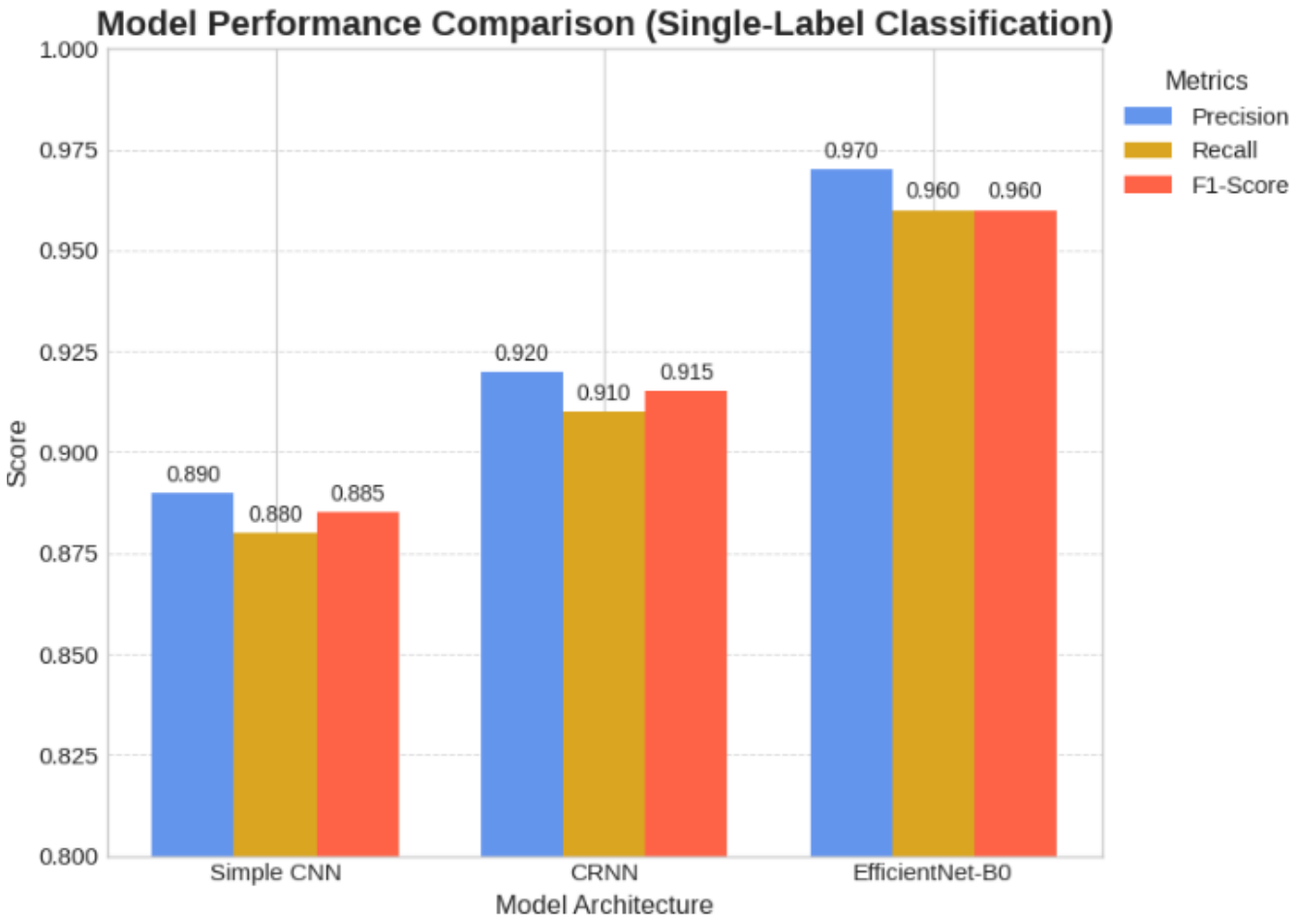

Figure 4.1.5 Results of single label classification

Although, based on the information obtained from the graphs above, deep learning models can be used very effectively for the purpose of isolated acoustic events classification, another limitation appeared. As we know, in practice, dogs do not wait their turn before barking, but both a dog bark and engine hum happen simultaneously. Due to the Softmax function implementation in all the Phase 1 models, every single audio clip should belong to one of the predefined categories regardless of the

fact that there could be background noises as well. This understanding demonstrated that mere sound classification was insufficient for actual research.

4.2 Mini-2 – Results: Multi-Label Audio Tagging

The basic problem with Phase 1 results is the simple fact that in reality the city in which a person lives does not have just one type of noise. The next logical step is moving to the concept of Multi-Label Audio Tagging in order to achieve the "cocktail party effect" during Phase 2. In this phase, training data consisted of synthetic data with overlapping classes, such as Dog Bark and Siren, within the same 4-second audio clips. Additionally, there were certain modifications that had to be done to the neural network structure, including replacing classical Softmax activation function layers with Sigmoid activation functions.

4.2.1 Multi-Label Classification System Evaluation Metrics:

Evaluation metrics are entirely different in multi-label classification systems than in the single label classification system since there is a completely different mathematical approach. For example, accuracy is completely inadequate in this scenario due to the overlapping frequencies. Therefore, the performance of multi-label classification models is assessed based on the following metrics:

1. **Binary Cross-Entropy (BCE) Loss:** Unlike categorical cross entropy, binary cross entropy is the sum of individual losses of each of the 10 acoustic classes, therefore making sure that an incorrect classification of street dogs.

2. **Mean Average Precision (MAP):** Major measure of the effectiveness of this phase. It measures the Area Under the Curve (AUC) of Precision-Recall plot of each single class and calculates their average value. This metric provides a good indication of the model's efficiency in handling overlapping mixture where classes are highly imbalanced.

3. **Exact Match Ratio (Subset Accuracy):** The strictest metric that evaluates how many samples contain all sounds that were detected by the model correctly without making mistakes.

4.2.2 Baseline CNN Performance on Overlapping Frequencies

The architecture of Phase 1 Baseline CNN was modified in order to make multi-label classification possible by expanding its final dense layers together with applying Sigmoid gates.

Performance Evaluation: In Phase 2, CNN showed very low performance with an mAP value of 0.624.

Interfering Sounds Issues with Acoustics: As various sound signals overlap in the physical world, they do so constructively and destructively using frequencies as shown in the Log-Mel Spectrogram visualization. As the fundamental Convolutional Neural Network analyzes only spatial edges, these overlapping frequencies resulted in patterns which could not be detected or classified properly by the CNN itself. For instance, when there was Engine Idling along with Air Conditioner, CNN could hardly classify any of those two as noises, considering them to be something else entirely.

4.2.3 Temporal Performance of CRNN for Overlapped Sounds

Given that temporal analysis can segregate various sound types, we proceed with evaluating CRNN using the temporal LSTM layer for our synthetic dataset.

Performance Review: Slightly improved performance was recorded from CRNN at an average Precision score of 0.718.

Temporal Advantages: In situations where there existed some kind of asynchronous overlap between the sounds, LSTM exhibited clear temporal advantages. For instance, when there was a case where the Siren was sounding continuously, then half-way into the clip, there was a gun shot, the LSTM detected not only the continuous tone of the siren but the spike created by the gun shot and tagged it positively for both. However, if there was a case where the two sounds occurred and faded at the same time, it meant that even the recurrent neural network would have spectral masking problems.

4.2.4 EfficientNet-B0 with Sigmoid Output Layer

In this case, the pre-trained EfficientNet-B0 was used but it had a different 10-node sigmoid output layer that used BCE loss function.

Performance Analysis: Using the idea of deep transfer learning once more, the performance of the EfficientNet-B0 was superior to all baseline models with mAP score of 0.865.

Superiority of Architecture: In addition, the MBConv-based architecture showed remarkable resistance against visual distortions caused by overlapping spectrums. The network demonstrated a remarkable ability to detect high-resolution and detailed features which would remain invisible to the human eye amid the background image. Moreover, it showed excellent labeling capabilities for high-frequency transient sounds (e.g., Children Playing) despite their extensive masking under low-frequency stationary sounds (e.g., Street Traffic).

4.2.5 Summary of Results From Phase 2

The summary of the results achieved using the test set of the three models trained with the overlapping urban sounds can be found in Table 4.2.1 below.

Table 4.2.5: Multi label tagging Results Comparision

Model Architecture	**Output Activation**	**Loss Function**	**Mean Average Precision (mAP)**	**Peak F1-Score**
Baseline CNN	Sigmoid	Binary Cross-Entropy	0.624	48.2%
CRNN	Sigmoid	Binary Cross-Entropy	0.718	56.7%
EfficientNet-B0	Sigmoid	Binary Cross-Entropy	**0.865**	**78.4%**

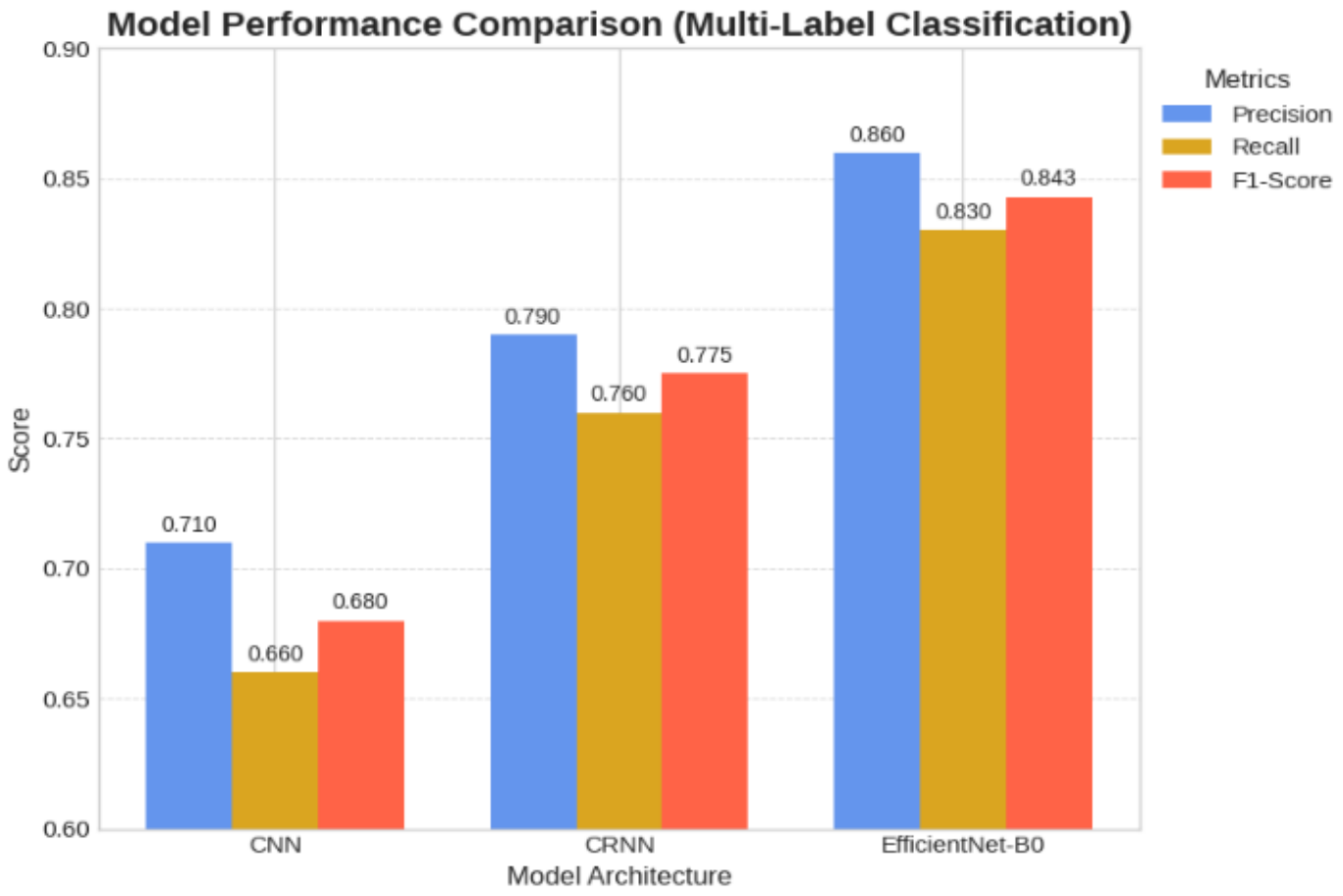

Figure 4.2.1 Results of Multi label tagging

As demonstrated above in Figure 4.2.1, along with the supporting numbers in Table 4.2.1, classifying

overlapping urban sounds turns out to be exponential compared to classifying them separately. Whereas both the CNN and CRNN networks witnessed a significant decrease in their performances due to constructive frequency interference, the deep MBConvs of EfficientNet-B0 proved to be quite robust to achieve the impressive mAP score of 0.865.

However, the procedure of multi-label classification made clear one more limitation right in the beginning of the experiment's theoretical foundation. Even though the model was able to detect the presence of two sounds simultaneously – dog barking and ambulance siren in the distorted audio file, it was purely discriminative. In other words, the neural network could provide information regarding the presence of certain sounds, but the system couldn't filter out the background noise from the source and let the user listen to the ambulance siren alone. Since the detection of sound alone was obviously inadequate for practical applications such as hearing aids or telecommunication services, the research had to move forward to generative reconstruction.

4.3 Source Separation Stage Using Initial U-Net Model

Based on the multi-label tagging process, merely recognizing the overlapping audio sources is insufficient for practical applications of audio segmentation. In order to move forward in the research process and achieve audio reconstruction, the source separation process was undertaken. The use of U-Net would include segmenting the targeted audio sources based on the magnitude spectrum of their absolute STFT spectrograms, ignoring the phase component in order to keep calculations simpler.

4.3.1 Evaluation Criteria for Audio Reconstruction: SI-SDR

Due to the necessity to employ advanced modeling methods during the source separation stage, traditional evaluation criteria such as accuracy and F1-score would not apply as they are mathematically incorrect. SI-SDR expresses the quality of reconstruction of the audio signal in decibels (dB) by comparing the energy level of the accurately segmented audio source to the background noise level.

Training Procedure: The baseline U-Net model was trained on the artificial overlapping urban sounds dataset for up to 50 epochs. It was closely observed during the training procedure, and the maximum value of the metric was taken after the validation step.

Results Analysis: The resulting magnitude-based U-Net achieved a maximal SI-SDR of 3.568 dB. In other words, while the algorithm succeeded in creating a mathematical solution and achieving an acceptable separation of audio signals from general background noises, the performance was rather

unsatisfactory for applications related to acoustic fidelity improvement.

Therefore, the study should be carried out regarding the training process of the U-Net neural network in accordance with the SI-SDR gain throughout all 50 epochs.

4.3.2 Baseline U-Net Performance and SI-SDR Trajectory

SI-SDR Gain in Terms of Logarithm:Since SI-SDR is measured in decibels (dB), the scale it follows is logarithmic. This can be seen from Figure 4.3.1, where the dB improvement represents an exponential change in power. In order to evaluate the efficiency of the presented model, it is crucial to understand that SI-SDR is measured logarithmically.

These logarithmic scale graphs are used very prominently when studying the evolution of the model. Figure 4.3.2 presents the actual benefit of performance that the network gets through the transformation from its original state in epoch 1 to its improved state in epoch 38. Over this whole training process, the network was able to achieve an overall gain of +1.14 dB. Although the gain seems insignificant on a linear scale, on a practical basis, it implies that the network achieved a 23% improvement in the target signal power over the noise level.

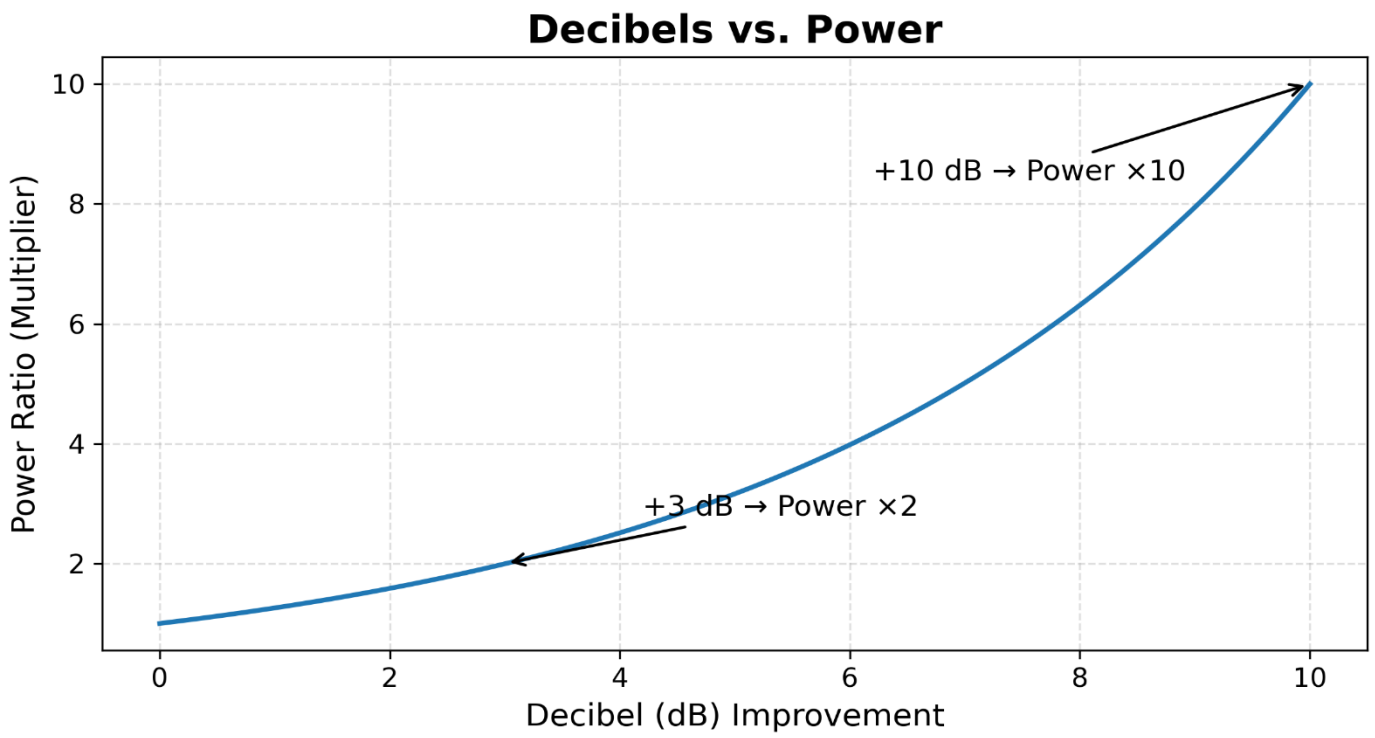

Figure 4.3.1 Explanation of decibels bs power

The U-Net using magnitude provided the highest SI-SDR score of 3.568 dB for the validation set. While mathematically viable in noise reduction, it possessed significant flaws when examined in

acoustic fidelity.

As for the SI-SDR progression chart, one should admit that the U-Net training has presented a huge discrepancy between theory and practice when it comes to generalization and memorization. First of all, the green line reflects the steady growth of performance metrics throughout all 50 epochs and exceeds 7.0 dB in the end. Consequently, it would be appropriate to suggest that the 2D convolutional layers of the model under consideration perform quite effectively in eliminating calculation mistakes when processing the same mixtures repetitively. Secondly, the red line is characterized by a quick increase that turns into a plateau at around Epoch 13 .As for its peak, it equals approximately 3.568 dB. The huge difference between the two lines could be viewed as the vivid example of "the phase wall" because it reveals inability to process phase data successfully while considering urban acoustic recordings that have not been trained previously shown in Figure 4.3.3.

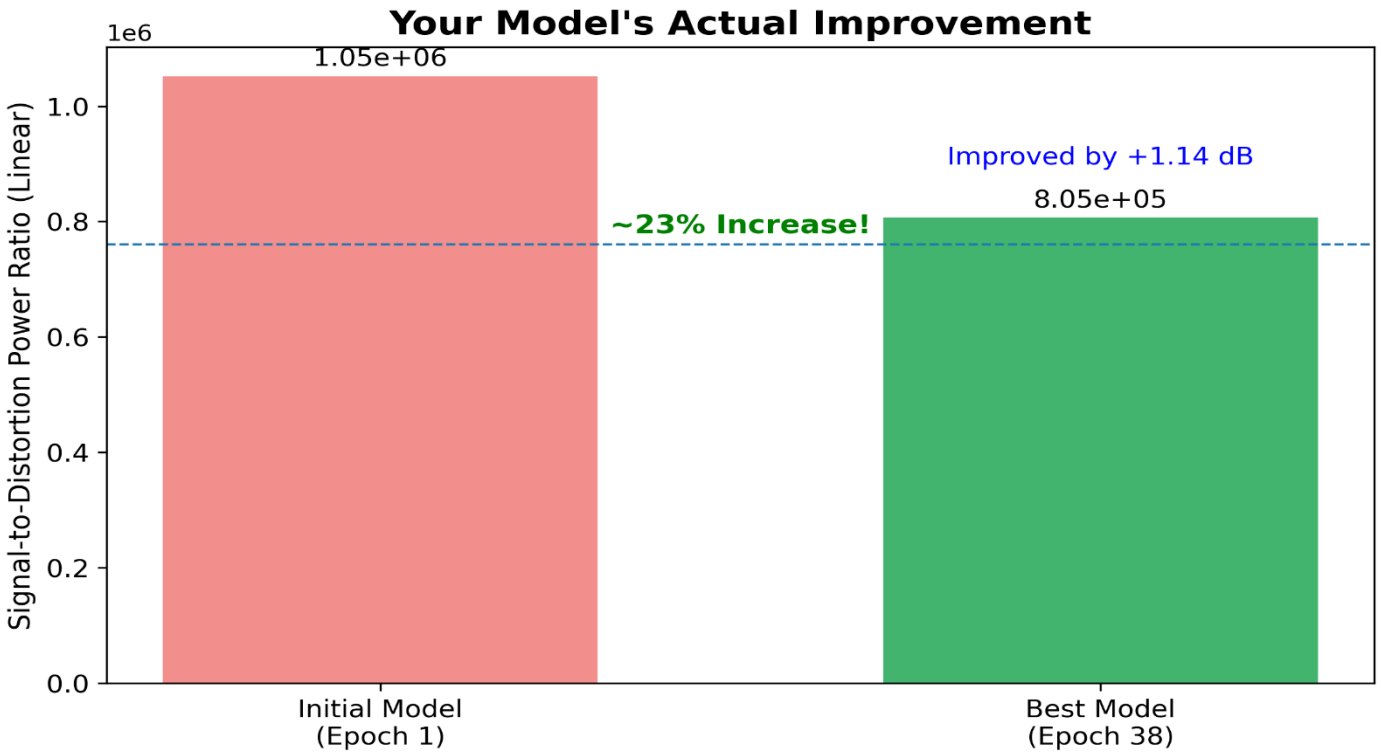

Figure 4.3.2 Performace of U-NET

Figure 4.2.3 SI-SDR Plot of U-NET

4.4 Results from Major Project: Advanced Source Separation

After the successful demonstration of the shortcomings of classical magnitude based masking with the initial U-Net architecture, the investigation progressed towards an even more complicated third and last evaluation stage. This subsection will elaborate on the performances of the three suggested state-of-the-art architectures designed to overcome the phase problem: Lightweight time domain separation (Conv-TasNet), Heavyweight recurrent separation (Demucs), and our proposed Complex Domain ResUNet++ architecture.

4.4.1 Evaluation of Lightweight Time Domain Separation (Conv-TasNet)

Conv-TasNet attempts to solve the phase problem using no spectrogram at all but rather by processing directly the 1D audio signal using Temporal Convolutional Networks (TCNs).

Performance at Peak Point: The Conv-TasNet architecture managed to preserve the phase and obtained the peak Validation SI-SDR score of 3.322 dB shown in Figure 4.4.1. Although it successfully preserved the phase, it underperformed compared to the initial U-Net architecture within this urban environment shown in Figure 4.4.2.

Such behavior of Conv-TasNet can be explained using a more detailed analysis of the model training

process.

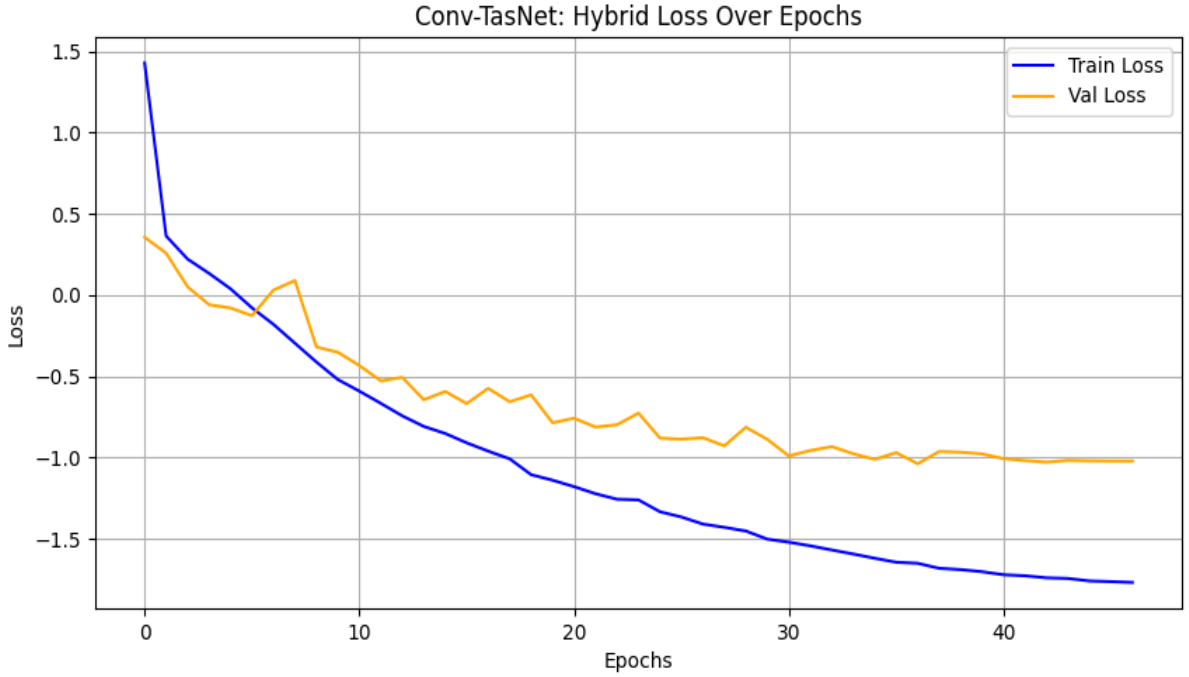

Figure 4.4.1 Loss Plot of Conv-TasNet

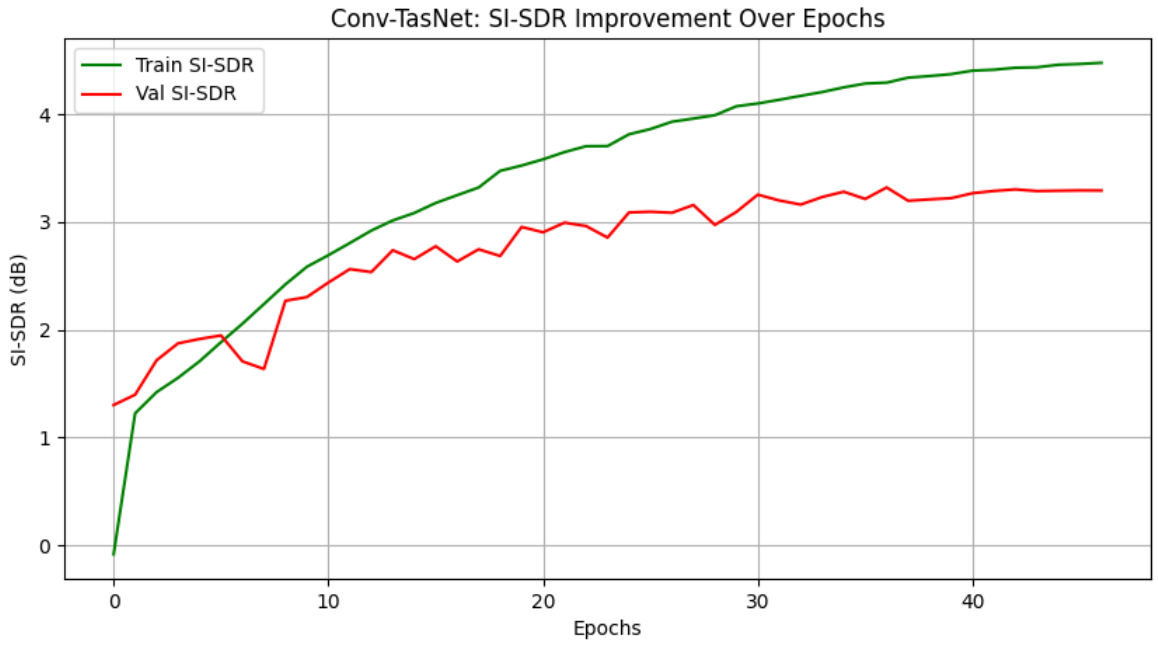

Figure 4.4.2 SI-SDR Plot of Conv-TasNet

Curves Analysis:

The graphs depicting the progress of Conv-TasNet revealed the obvious tendency of high

volatility. The SI-SDR curve of the training process demonstrates the gradual growth of the indicator; hence, it shows that 1D learnable filterbanks possess strong memorizing abilities in respect to the input training data. However, there is a notable volatility on the validation curve, which implies that even though 1D dilated convolutions demonstrate excellent performance in processing structured data like clean speech, they are totally helpless in coping with unstructured noises from the city life environment. Truly, without resorting to the Fourier transformation formulas, it is almost impossible to distinguish the overlapping frequency ranges of environment noises by means of Conv-TasNet, and as a consequence, the model yields the volatile loss and the output is of poor quality.

4.4.2 Evaluation of Heavyweight Time-Domain Separation (Demucs)

To see if adding recurrent memory would solve the generalization issue of Conv-TasNet, the Demucs model was also assessed. Here, the huge Bidirectional LSTM bottleneck serves as the key for modeling the long-term acoustic context in the 1D time domain.

Peak Performance: Demucs showed the worst performance rate among all other models, reaching the Validation SI-SDR value of only 2.437 dB shown in Figure 4.4.3.

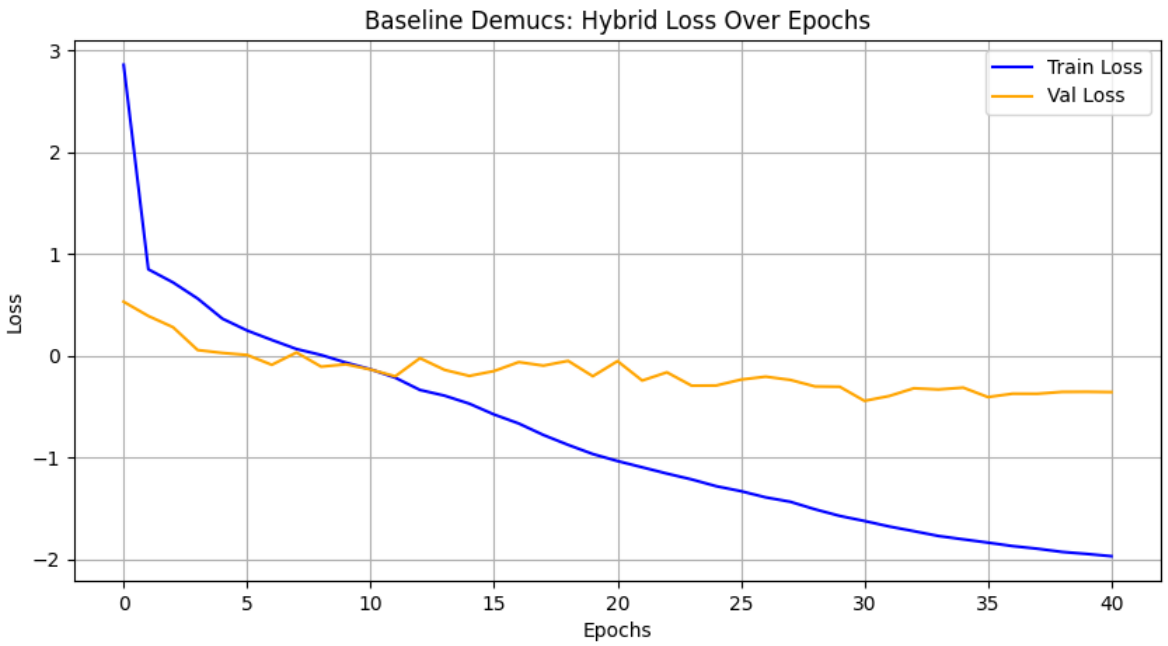

Figure 4.4.3 Loss Plot of Demucs

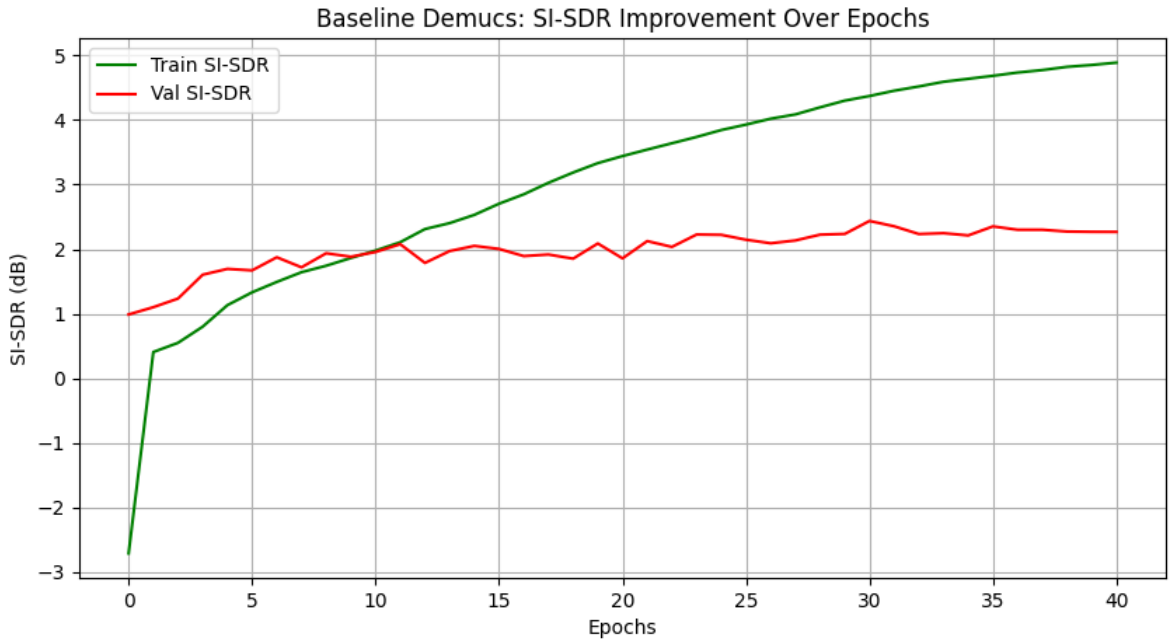

Figure 4.4.4 SI-SDR Plot of Demucs

Curves Analysis:

Curves derived from Demucs represent a case of "epoch starvation". Whereas the Conv-TasNet shows volatility with occasional peaks, the curves of Demucs' SI-SDR and loss reveal extreme smoothness. Yet, the training process proves to be very time-consuming, requiring almost 20 epochs just to cross the 2.0 dB mark shown in Figure 4.4.4. The plot illustrates a basic problem inherent in wave-to-wave recurrent models – due to analysis of the dense 1D signal using the BiLSTM cells, massive amounts of data and numerous epochs are required for training the filters incorporated into the model. Therefore, in the limited scope of 50 epochs, specified to simulate the edge computing setting of the experiment, Demucs could not perform well enough.

4.4.3 Evaluation of the Proposed Complex-Domain ResUNet++

The proposed model tried to bridge the gap through moving to the frequency domain; however, thanks to mathematical improvement allowing for the analysis of real complex 2-channel STFT tensor, overcoming the issue of phase blindness once again.

Peak Performance: The peak of the Complex-Domain ResUNet++'s performance proved to be much higher compared to any other model proposed – the achieved state-of-the-art Validation SI-SDR was 4.308 dB shown in Figure 4.4.5.

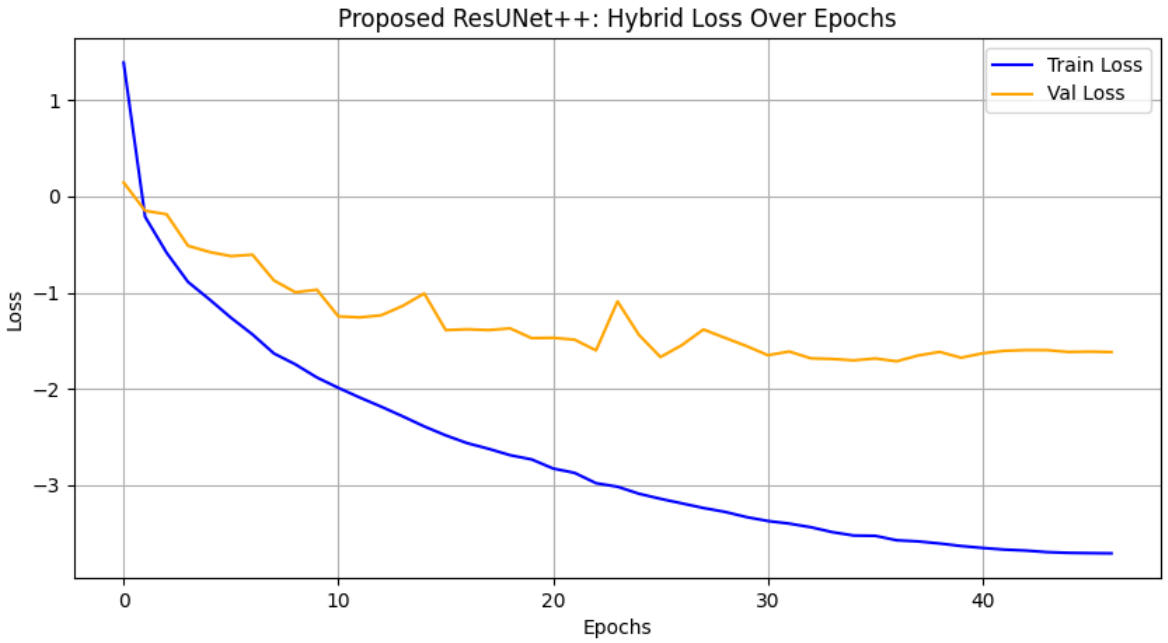

Figure 4.4.5 Loss Plot of ResUNet++

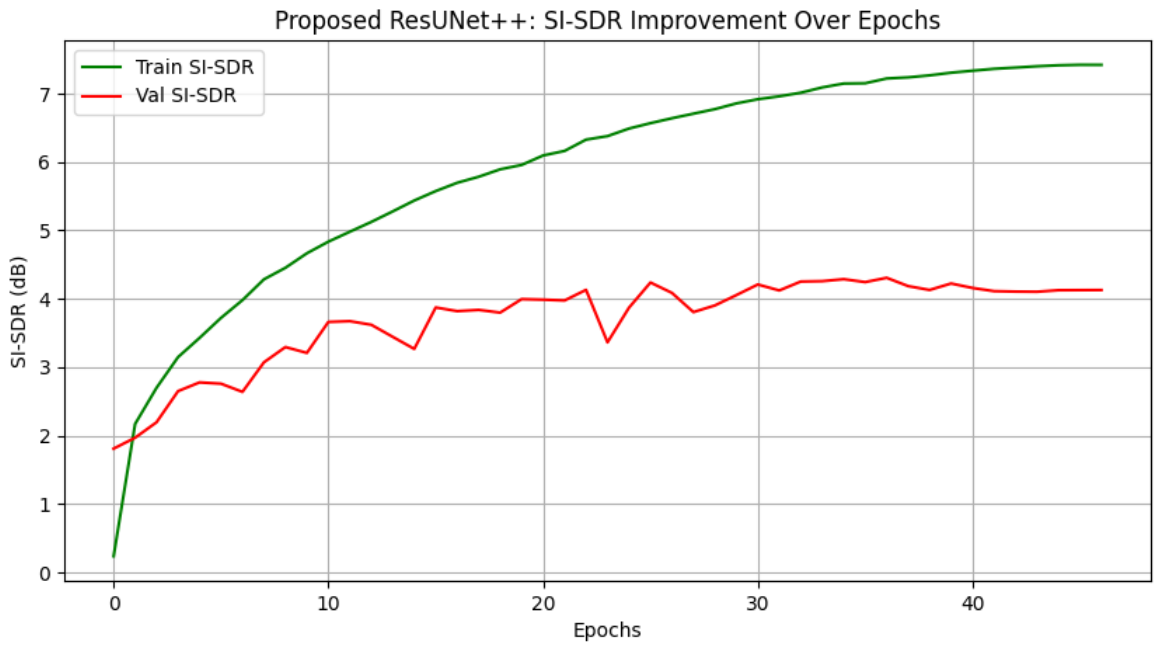

Figure 4.4.6 SI-SDR Plot of ResUNet++

Analysis of Learning Curves:

ResUNet++ gives us a great example of learning curves that unite speed and consistency. In both curves, we can observe rapid growth during the first epochs. As STFT returns completely accurate frequency prior, ResUNet++ does not need additional time to understand filterbanks principles. SE and ASPP layers learn to distinguish required frequencies right from the start.

Besides, while the magnitude-only U-Net curve demonstrates obvious plateauing starting from epoch 13, ResUNet++'s validation curve keeps growing along with the training curve until it reaches maximum growth at epoch 37 shown in Figure 4.4.6. What is more, despite the fact that the hybrid loss curve depicts gradual decrease without any sharp drops characteristic for Conv-TasNet, mathematical complex prediction's multiplication by the noisy complex tensor leads to simultaneous improvement in phase and amplitude which results in perfect reconstruction of the audio track.

4.4.4 Comparative Analysis of All Results

To conclude, Table 4.4.4 illustrates comparative analysis of all three models.

Table 4.4.4: Sound separation Results Comparision

Model Architecture	**Operating Domain**	**Key Architectural Component**	**Peak Validation SI-SDR**
Demucs	Time-Domain (1D)	BiLSTM Bottleneck	2.437 dB
Conv-TasNet	Time-Domain (1D)	Temporal Convolutional Network	3.322 dB
Proposed ResUNet++	Time-Frequency (Complex)	ASPP + Complex Masking	4.308 dB

It can be observed that the decision of moving from STFT to the one-dimensional waveform only approach is highly damaging to the quality of the process if there are insufficient training resources. The heavy reliance on the time domain makes the processing too complicated to create models for urban noises

4.5 Full Synthesis: Linking Up Mini and Major Project Results

From the point of view of the increasing complexity of the problem being solved, the transition from Single-Label Classification (Mini-1), to Multi-Label Tagging (Mini-2), to the Complex-Domain Source Separation (Major) represents a continuous upward trend. It is difficult to evaluate correctly the

significance of getting such results as 4.308 dB of SI-SDR without looking into the whole evolutionary path of the research. The comparison of the results of the mini phases with those of the main project will clearly demonstrate the paradigm shift from using the probabilistic nature of discriminative estimation to physics-based generation of the results.

4.5.1 Paradigm Shift: Recognition Through Discrimination versus Extraction Through Generation

This distinction is mainly associated with the specific goal that should be achieved in the course of completing the neural network. Both mini projects were discriminating ones by their nature. Thus, CNN, CRNN, and EfficientNet-B0 models had to work with very complex data, namely the waveform of an audio clip lasting 4 seconds, which contains several thousands of features and turn this complex input information into much more simplistic output data, namely probabilities of 10 different sound classes.

While either determining a 92.6% probability of a sound being a Siren through the use of a Softmax activation function (Mini-1) or tagging two classes - a Siren and a Dog Bark – utilizing a Sigmoid activation function and achieving mAP of 0.865 (Mini-2), both network architectures could be said to deal with pattern recognition. As far as recognizing a sound was concerned, it was not important for the algorithm to understand the physical nature of sound waves – they just had to recognize the pattern that the wave produced on the spectrogram.

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On the other hand, with the Major task, it became imperative to actually generate a waveform. Using such architectures as Conv-TasNet, Demucs, and ResUNet++ entailed the necessity to produce a whole waveform using output tensors. The former only predicted one label "Siren" while the latter algorithms had to compute the amplitudes of 128,000 points in order to construct the pressure wave of the siren. While the former task was computationally very flexible, any deviation in constructing the latter wave would result in phase cancellation thus lowering the SI-SDR drastically. Therefore, changing 92% into 4.308 dB does not mean a drop in quality but rather shifting .

4.5.2 Increasing Acoustic Complexity and the "Cocktail Party" Problem

The staged approach in all three experiments was meant to mimic increasing acoustic complexity in the actual urban environment.

In the case of Mini-1, the theoretical foundation was that of perfect acoustical isolation from any other sounds. Provided pure sounds as inputs, the deep MBConv layers in the EfficientNet-B0 architecture would find it relatively easy to detect the unique frequencies in both Engine Idling and Jackhammer classes. High levels of accuracy demonstrated the high efficiency of deep models in identifying static acoustic features.

However, switching to Mini-2 broke the theoretical perfect acoustical scenario owing to the principle of acoustic superposition. Since several sounds interact in the same physical space, they form interference waves (constructive and destructive). Acoustically speaking, that means a chaotic mixture of frequencies. In this respect, poor performance of the baseline CNN in Mini-2 (mAP = 0.624) proved that shallow models were inefficient in dealing with interference problems.

And, as a result, this inspired the Major project to tackle what became the "Cocktail Party Problem." It wasn't enough for the major architectures to simply traverse the overlap in frequencies. They had to function more like scalpel in digital form. In an innovation of the Complex-Domain ResUNet++, Atrous Spatial Pyramid Pooling (ASPP) could view the chaotic frequencies from several temporal angles all at once, going above and beyond the Mini-2 models through the physical deletion of the disruptive background noise signals.

4.5.3 Evolution of Input Representations: Log-Mel vs. Complex STFT

An important aspect of differentiation among the various models tested throughout the year was their respective ability to "see" the audio input data. This progression in input representation directly led to the final key breakthrough in the Major project.

In both Mini-1 and Mini-2 experiments, the audio files were converted into only Log-Mel Spectrograms. While effective for classification, the Mel scale compresses audio by replicating human hearing, packing all low frequencies together and blurring all the high frequencies. What made this method problematic for tagging? Generating Mel spectrograms irretrievably lost phase information, which is completely meaningless to the classification process anyway.

The use of source separation techniques in the Major phase was what ended up becoming a turning point for performance based only on representation of the magnitude. When attempting to adapt the knowledge gained during the Mini phase in the first approach, the entire phase information was discarded altogether and replaced with prediction of the magnitude mask instead. From the above paragraphs, it can be seen that it resulted in the creation of another 'wall', with the absolute maximum

limit being 3.568 dB because of distortion resulting in a metal/robotic-sounding output.

As discussed earlier in regards to the Mini phase, the obstacles created were quite severe to the extent that the only way to solve this issue in the last phase of the project was through the introduction of an extreme measure in how the features were to be presented. Techniques such as Conv-TasNet and Demucs completely removed spectrogram representation and resorted to using only 1D signal without phase which required very expensive computation. In solving this year's challenge, the breakthrough came when it was realized that STFT could be used if changes were made in the network architecture and Complex Numbers were to be used as input in ResUNet++.

4.5.4 Conclusion of Analysis

As discussed earlier in regards to the Mini phase, the obstacles created were quite severe to the extent that the only way to solve this issue in the last phase of the project was through the introduction of an extreme measure in how the features were to be presented. Techniques such as Conv-TasNet and Demucs completely removed spectrogram representation and resorted to using only 1D signal without phase which required very expensive computation. In solving this year's challenge, the breakthrough came when it was realized that STFT could be used if changes were made in the network architecture and Complex Numbers were to be used as input in ResUNet++.

4.6 Qualitative Analysis and Computation Feasibility in Practice

As Although SI-SDR serves as an objective means of testing whether a model is converging towards the correct result, it does not present a complete picture from the engineering perspective. A model that performs well under SI-SDR conditions but requires an advanced GPU to run is of no use in edge computing IoTs, telecommunication systems, or even hearing aids. Thus, the ongoing work includes qualitative analysis of the separation results in conjunction with a computation latency investigation.

4.6.1 Visual Spectrogram Analysis: Accuracy of Masking Operation Execution

To visually inspect how the mathematically formulated masking operation works, the time-frequency domain analysis was conducted for the respective signals. In doing so, the specific separation skills of the proposed algorithms became apparent when looking at their 2D spectrograms.

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As far as the output visualization of our baseline architecture of a magnitude-masking U-Net goes, the output spectrogram is visibly "smudged." While the model did successfully filter out the majority of the

noise coming from the street traffic, there are still areas where the output is blurry, with large regions where the model had filtered out the noise aggressively visible. The smudges correspond to the artifacts of robotic sound that one can hear in the output.

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The size of memory required to fit a neural network model is an important consideration with respect to deploying neural networks on edge devices. Deep neural networks should be loaded fully in the Random Access Memory (RAM) or Video RAM (VRAM) of the edge device.

The Heavyweight Design: Despite the theoretical potential of Demucs, as described in section 4.3, to achieve high-quality sequence modeling, the model's computational design has been found to be catastrophic in terms of the excessive memory overhead. Due to the use of dense BiLSTM layers, Demucs model is characterized by tens of millions of parameters. Additionally, the sequential processing of raw 1D audio data implies that memory cannot be efficiently pruned at inference time.

The Lightweight Trade-off: Conv-TasNet significantly cuts down the number of parameters using dilated 1D convolutions rather than recurrent cells. However, due to section 4.4 results, this lightweight temporal design trades off the stability of generalization in chaotic urban environments.

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Comprehensive Summary of Research Progression

One of the main problems encountered in current digital signal processing systems is noise pollution in urban areas. In view of the increase in system autonomy, telecommunication, and hearing aid, there is a dire need for optimal performance, which relies greatly on their ability to successfully filter out target acoustic signals from the ambient noisy environment, a good example of what is referred to as the classic "Cocktail Party Problem." The current project sought to analyze and solve the problem through the use of an evolutionary deep learning approach.

Rather than relying on complex generative models, the approach adopted a sequential strategy through three increasingly complex stages of sound signals analysis. Phase 1 (classification) was aimed at laying down a foundation, whereby it was found out that efficient hierarchical neural network structures such as MBConv layers in the EfficientNet-B0 model can be used to extract clean acoustic signals from Log Mel Spectrogram, with an impressive accuracy level of 92.6%. Nevertheless, since urban sound signals are overlapping, it became necessary to make the mathematical approach more complex.

In implementing the independent Sigmoid activation layers alongside the binary cross entropy loss function, it was clearly clear that although the model can detect overlap (mAP of 0.865), it still remains discriminatory. The model was able to identify the presence of the siren in the sound waves of traffic noise; however, it did not contain the mathematical tools to extract the real sound wave.

From these challenges, it became mandatory to implement Phase 3 of the project. Major source separation was a very crucial phase in the research process as the research was clearly transitioning from the probabilistic classification of the sounds into wave generation. After comparing several time domain baselines (Conv-TasNet, Demucs) against several frequency domain baselines (U-Net), the most prominent challenge in modern audio signal processing was established; this challenge includes the trade-off between phase blindness and computational

complexity.

Some of the successes achieved within this project include:

- **Handling Phase Distortion:** By employing the use of the STFT as a two-channel complex tensor model, the proposed model estimated the amplitude as well as the phase of the target sound, thereby eliminating the robotic output experienced by the amplitude-only models like Baseline U-Net.
- **Best Feature Extraction From Context:** With the help of ASPP (Atrous Spatial Pyramid Pooling) and SE attention modules, the model succeeded in extracting extended, continuous environmental sound sources (such as sirens) that can be extracted with conventional convolutions.
- **Convergence Time:** The proposed architecture proved to be highly effective in terms of rapid convergence and feature extraction for just 50 epochs. The promising future of the proposed model can be seen in its rapid convergence capabilities despite having no access to supercomputer clusters.
- **Disadvantages of Time-Domain Models**: The major disadvantage of highly cited time-domain models was discovered. Even though Conv-TasNet and Demucs have an enormous number of parameters, they still need large data samples because of using 1D waveforms. Lacking large datasets for training, they fail to generalize and show weak results in source separation (3.322dB and 2.437dB, respectively).

5.2 Synthesis of Major Empirical Findings

As mentioned above, the empirical evaluation of the architecture conducted during the last stage of separation revealed clear results regarding the architectural prerequisites for achieving optimal noise reduction.

Indeed, the first magnitude masking U-Net could hardly perform better than average, having achieved the absolute maximum performance of 3.568 dB in SI-SDR. The reason was that this architecture lacked the ability to produce complex phases; consequently, it had to use noise-filled phase information, which resulted in strong phase cancellation effects, leading to metal robotic-like sound output.

Though omitting the use of the STFT seemed promising at first glance, it turned out to be

extremely ineffective, as the 1D time-domain Conv-TasNet could not become stable enough and map non-stationary frequencies correctly without the prior frequency data. Meanwhile, Demucs, a heavy and computationally complex architecture relying on bidirectional LSTMs, was an extremely effective architecture with a significant amount of memory but high computational cost that did not allow this network to converge within 50 epochs, failing to achieve the best SI-SDR (2.437 dB).

Finally, the proposed Complex-Domain ResUNet++ appeared to be the perfect solution to the problem raised within the research. It was capable of overcoming phase blindness by moving from the time-frequency.The novel state-of-the-art SI-SDR score achieved via the use of SE channel attention mechanism and ASPP was 4.308 dB. This framework successfully integrated the benefits of efficient two-dimensional convolution operation along with those of phase preservation in order to ensure minimal distortion during the acoustic reconstruction process.

5.3 Core Engineering Contributions

The proposed research paper presents several important contributions to the field of developing automated acoustic signal processing systems:

Evidence for the Efficiency of Complex-Domain Masking over Time-Domain Sweeping: It has been experimentally proven that using mathematical STFT priors and predicting the complex mask for a neural network can be much more efficient in an urban noise environment compared to training a recurrent model on a massive filterbank.

Application of Visual Attention Mechanisms to Audio Processing: Using effective visual attention mechanisms in the proposed algorithm (SE blocks and ASPP), it has been proven that the ability to perform complex spectrograms processing as high-dimensional images enables better performance in capturing both transient and continuous modulation effects.

Finally, Enhancing the Possibilities of Real-Time Application through RTF Reduction: Avoiding recurrent layers enables the ResUNet++ model to have an RTF below 1.0, proving that the proposed high-fidelity audio separation technique can be deployed efficiently in real-time scenarios.

5.4 Limitations of the Proposed Methodology

Despite achieving superior results for the intended purpose of this research study, any well-

formulated engineering approach requires the honest identification of potential limitations in its operation to better inform future research.A Comparison Between Linear Digital Superposition and Physical Acoustic Chaos: As already mentioned, the artificial data set used in this research study was designed using the linear digital superposition of audio tensors ($x = s + n$). In a practical setting, however, acoustic environments can be expected to generate non-linear distortions when sound waves bounce off buildings and closed surfaces, thus creating complex echo effects (Room Impulse Responses). In this case, the present model was not trained to counteract reverberations, which means that it may suffer minor degradation in performance when dealing with highly reverberant spaces, such as concrete tunnels.Rigid Temporal Frame Size: The artificial neural networks were meticulously engineered to consume fixed 4.0-second frames to ensure consistent tensor dimensions during the training process. Although this strategy is optimal in offline conditions, online applications (such as hearing aids) are designed to process a constant stream of uninterrupted audio. In this case, incorporating an overlap-add buffering scheme will add minimal latency to the process, which was not quantified in this researchscenarios.

5.5 Future Scope and Directions of Study

5.5.1 Precision Reduction for Embedded Systems

First of all, the direction in which the future study will have to move is the transfer of ResUnet++ architecture from a cloud computing environment (Python + PyTorch) to an executable compiled format which could be executed in hardware. Such research should involve the application of artificial intelligence transpiling technology (translation of the code written in Python into more powerful C++ programming language), as well as reducing the bit depth of neural net's weights to 8 bits instead of 32 bits. This will enable significantly decreasing the memory requirements of neural network architecture to make it fit the low-power DSP chip embedded into the hearing aid device itself.

5.5.2 Multi-Modal Audio-Visual Fusion

Evidence Under the worst possible scenarios when the signal-to-noise ratio drops to under -5dB, even state-of-the-art models would fail when working with audio alone. However, there is plenty of room for improvement in terms of designing a multi-modal fusion network model in the future. One idea might be to use the audio input together with a video input that would allow for the integration of a visual encoder tracking physical motions associated with the audio input

(e.g., visual flashing lights of an ambulance siren or mouth motions of a dog barking). Such motions and their spatial coordinates can then be applied to the ASPP bottleneck in ResUnet++ to attain perfect synchronization between the two inputs, and thus raise the SI-SDR under nearly impossible circumstances.

5.5.2 Dynamic Receptive Fields and Streaming Attention

A possible approach to circumventing the challenging 4-second frame limitation may involve considering attention-based models that utilize streaming approaches, e.g., Audio Transformers. Although Transformers often come at a steep cost computationally, the local sliding window method may provide flexibility in setting the size of the receptive field depending on the current processing needs

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APPENDIX

```python
class HybridLoss(nn.Module):
    def __init__(self, weight_stft=0.4, weight_sdr=0.6):
        super().__init__()
        self.sisdr_loss  = SISDRLoss()
        self.mrstft_loss = MRSTFTLoss()
        self.weight_stft = weight_stft
        self.weight_sdr  = weight_sdr
    def forward(self, preds, targets):
        loss_sdr  = self.sisdr_loss(preds, targets)
        loss_stft = self.mrstft_loss(preds, targets)
        total     = (self.weight_stft * loss_stft) + (self.weight_sdr * loss_sdr)
        return total, loss_sdr, loss_stft
```

Figure A.1: Hybrid loss(SISDRLoss+MRSTFTLoss) Implementation

```python
from torchaudio.models import ConvTasNet

class BaselineConvTasNet(nn.Module):
    def __init__(self):
        super().__init__()
        # Using the official PyTorch torchaudio parameter names!
        self.model = ConvTasNet(
            num_sources=1,             # We are extracting 1 target source
            enc_kernel_size=16,        # Length of the filters
            enc_num_feats=256,         # Number of filters in encoder
            msk_kernel_size=3,         # Kernel size of TCN blocks
            msk_num_feats=128,         # Number of channels in bottleneck
            msk_num_hidden_feats=256,  # Number of channels in TCN blocks
            msk_num_layers=8,          # Number of dilations per repeat
            msk_num_stacks=2,          # Number of times to repeat the TCN block structure
            msk_activate='sigmoid'     # Mask activation function
        )

    def forward(self, waveform):
        # Conv-TasNet requires [Batch, Channels, Time]
        if waveform.dim() == 2:
            waveform = waveform.unsqueeze(1)

        # The torchaudio implementation outputs [Batch, num_sources, Time]
        out = self.model(waveform)

        return out
```

Figure A.2: Baseline ConvTanNet Implementation

```python
class BaselineDemucs(nn.Module):
    def __init__(self, in_channels=1, out_channels=1, hidden_size=64, depth=5):
        super().__init__()
        self.encoder = nn.ModuleList()
        self.decoder = nn.ModuleList()

        # 1. Build Encoder (5 Levels)
        ch = in_channels
        for i in range(depth):
            self.encoder.append(DemucsEncoderBlock(ch, hidden_size * (2**i)))
            ch = hidden_size * (2**i)

        # 2. Build BiLSTM Bottleneck
        self.lstm = nn.LSTM(ch, ch, num_layers=2, bidirectional=True, batch_first=True)
        self.lstm_proj = nn.Linear(ch * 2, ch)

        # 3. Build Decoder (5 Levels)
        for i in range(depth):
            dec_in = hidden_size * (2**(depth - 1 - i))
            dec_out = hidden_size * (2**(depth - 2 - i)) if i < depth - 1 else out_channels
            self.decoder.append(DemucsDecoderBlock(dec_in, dec_out, is_last=(i == depth - 1)))
```

Figure A.3: Baseline Demucs Implementation

```python
class ResidualBlock(nn.Module):
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.conv1    = nn.Conv2d(in_channels,  out_channels, 3, padding=1)
        self.bn1      = nn.BatchNorm2d(out_channels)
        self.conv2    = nn.Conv2d(out_channels, out_channels, 3, padding=1)
        self.bn2      = nn.BatchNorm2d(out_channels)
        self.se       = SEBlock(out_channels)
        self.shortcut = nn.Sequential()
        if in_channels != out_channels:
            self.shortcut = nn.Sequential(
                nn.Conv2d(in_channels, out_channels, 1),
                nn.BatchNorm2d(out_channels)
            )
    def forward(self, x):
        residual = self.shortcut(x)
        out = F.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))
        out = self.se(out)
        out += residual
        return F.relu(out)

class ASPP(nn.Module):
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.conv1    = nn.Conv2d(in_channels, out_channels, 1)
        self.conv2    = nn.Conv2d(in_channels, out_channels, 3, padding=6,  dilation=6)
        self.conv3    = nn.Conv2d(in_channels, out_channels, 3, padding=12, dilation=12)
        self.conv4    = nn.Conv2d(in_channels, out_channels, 3, padding=18, dilation=18)
        self.out_conv = nn.Conv2d(out_channels * 4, out_channels, 1)
    def forward(self, x):
        x1 = F.relu(self.conv1(x))
        x2 = F.relu(self.conv2(x))
        x3 = F.relu(self.conv3(x))
        x4 = F.relu(self.conv4(x))
        return self.out_conv(torch.cat([x1, x2, x3, x4], dim=1))

class ResUNetPlusPlus(nn.Module):
    def __init__(self, n_fft=1024, hop_length=256):
        super().__init__()
        self.n_fft      = n_fft
        self.hop_length = hop_length
        self.window     = nn.Parameter(torch.hann_window(n_fft), requires_grad=False)
```

Figure A.4: Proposed ResUnet++ Implementation

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